Paper versus reproduction: all 16 recognition cases

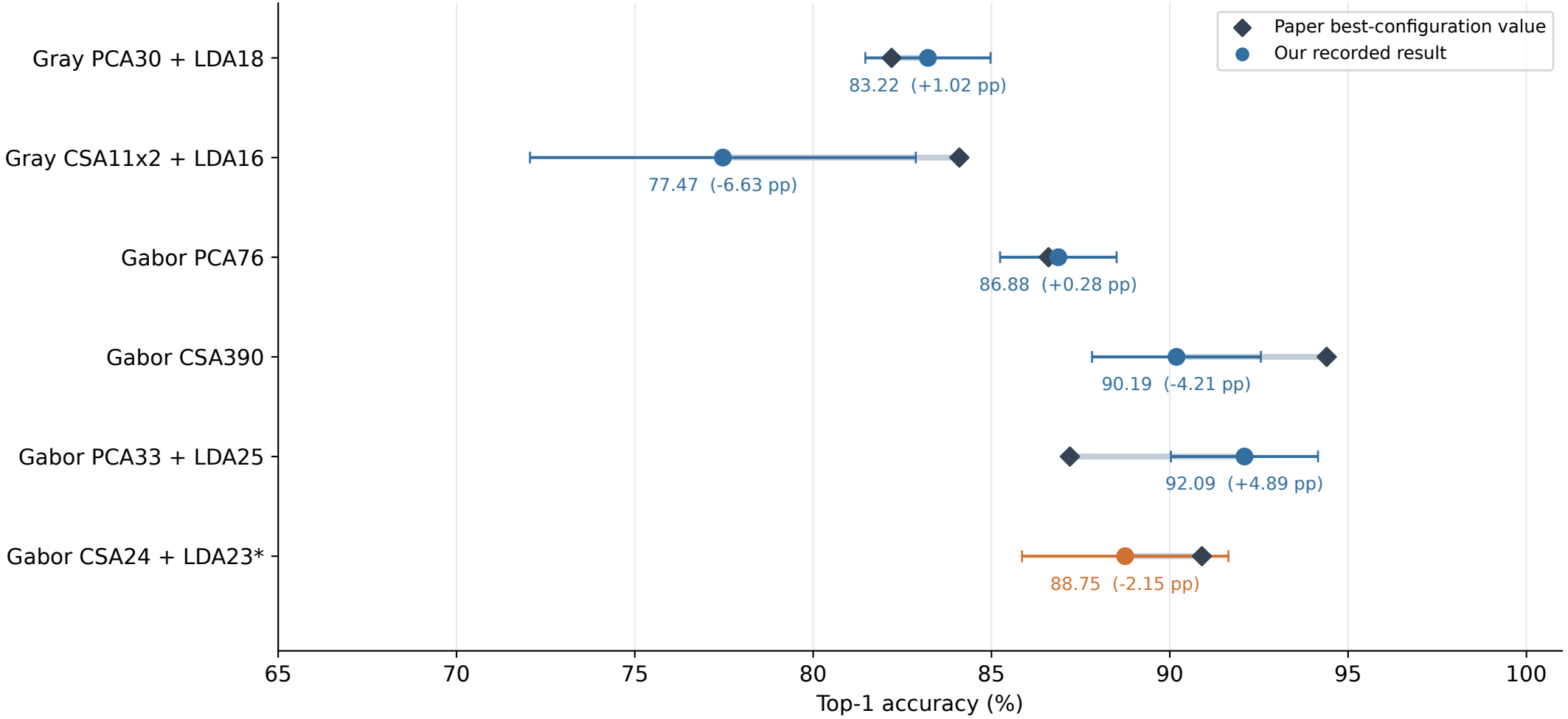
Full reproduction within the instructor-approved three-dataset scope; * marks the ORL context extension.

Dataset / setting	Method	Our accuracy (%)	Paper (%)	Difference (pp)
ORL	Gray PCA30 + LDA18	83.22 +/- 1.76	82.20	+1.02
ORL	Gray CSA11x2 + LDA16	77.47 +/- 5.41	84.10	-6.63
ORL	Gabor PCA76	86.88 +/- 1.63	86.60	+0.28
ORL	Gabor CSA390	90.19 +/- 2.37	94.40	-4.21
ORL	Gabor PCA33 + LDA25	92.09 +/- 2.06	87.20	+4.89
ORL	Gabor CSA24 + LDA23*	88.75 +/- 2.89	90.90	-2.15
MNIST G30	PCA60	87.52 +/- 0.40	88.50	-0.98
MNIST G30	CSA810	90.34 +/- 0.15	92.80	-2.46
MNIST G40	PCA50	88.58 +/- 0.37	90.50	-1.92
MNIST G40	CSA975	91.04 +/- 0.25	92.40	-1.36
FERET q=50	PCA45 + LDA25	89.08	84.60	+4.48
FERET q=50	CSA10x4 + LDA35	88.73	88.20	+0.53
FERET q=100	PCA95 + LDA40	90.87	90.60	+0.27
FERET q=100	CSA11x5 + LDA44	94.25	92.20	+2.05
FERET q=200	PCA112 + LDA69	93.87	93.40	+0.47
FERET q=200	CSA13x7 + LDA62	95.38	95.20	+0.18

ORL: ten splits; MNIST: three splits; error values are sample SD, not confidence intervals. FERET: one fixed identity partition, no split SD. ORL* uses HOSVD/context; other rows retain their reference protocols. Paper repetition/selection details are unspecified. Paper values: Xu et al., IEEE TCSVT (2008), Tables I/II/IV, doi:10.1109/TCSVT.2007.903317.

ORL: six paper-dimension recognition cases

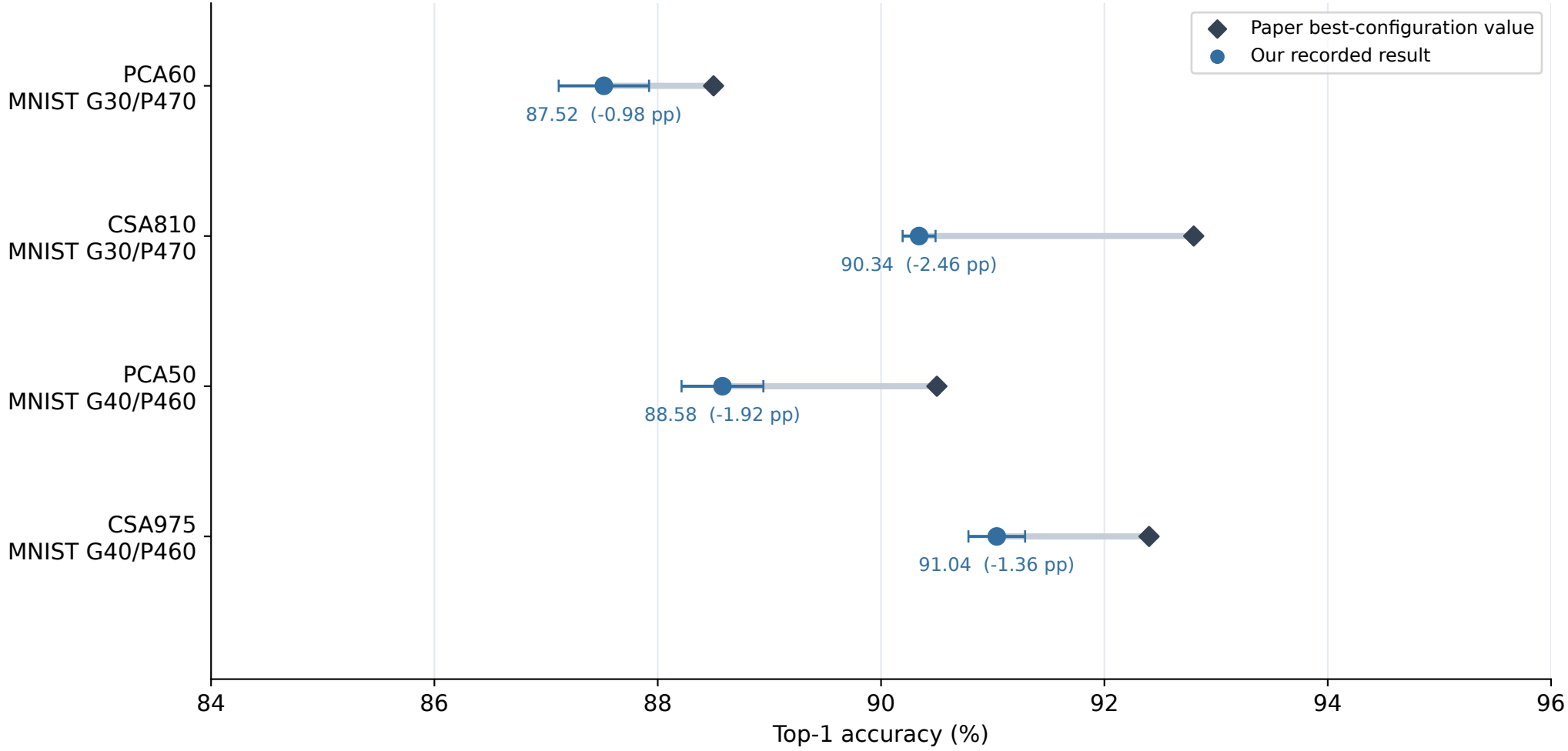
Diamonds: paper values. Circles: our means. Horizontal whiskers: sample SD.



ORL* is the current HOSVD/context extension at 24->23 dimensions, not a common input with every other ORL row. MNIST uses the original three-seed references. Best paper configurations and our repeated means have different reporting conventions. Paper values: Xu et al., IEEE TCSVT (2008), Tables I/II/IV, doi:10.1109/TCSVT.2007.903317.

MNIST: both gallery sizes, both representations

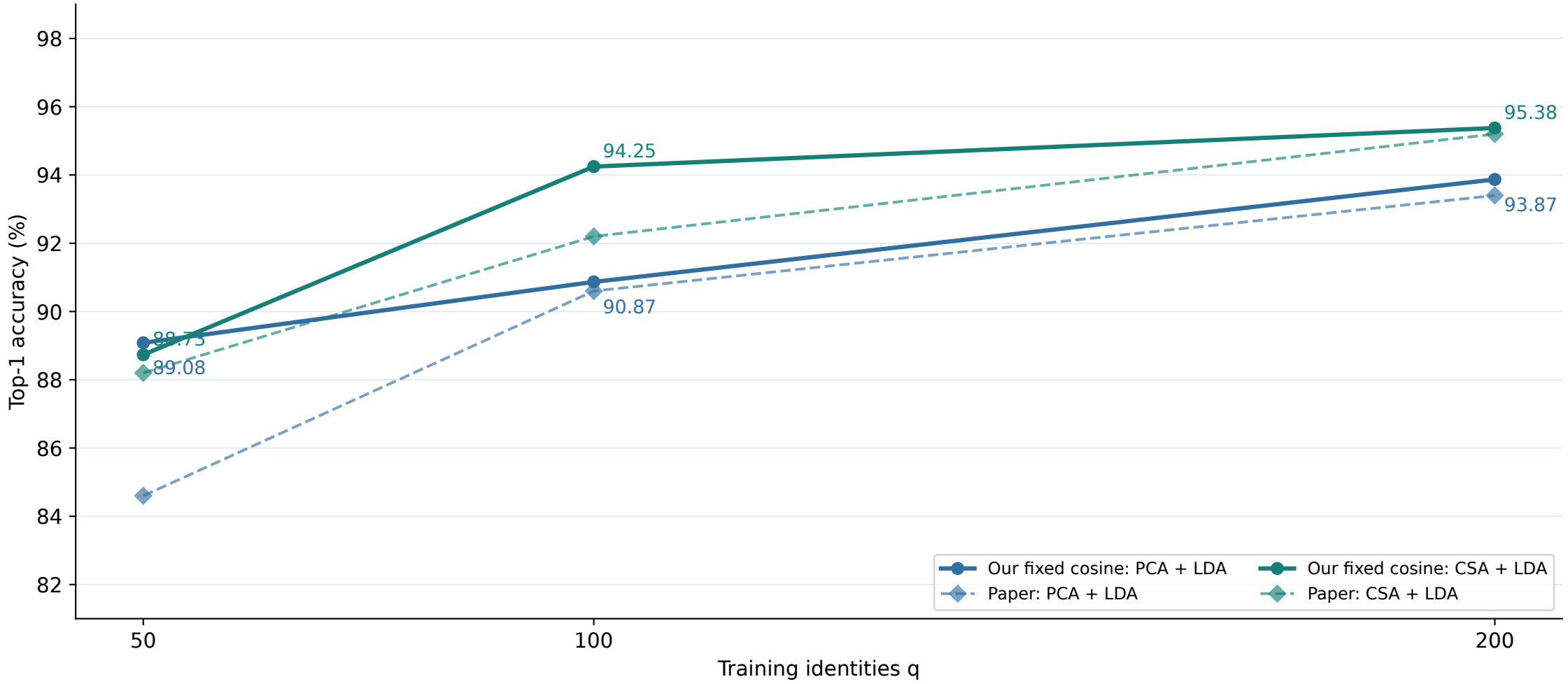
Diamonds: paper values. Circles: our means. Horizontal whiskers: sample SD.



ORL* is the current HOSVD/context extension at 24->23 dimensions, not a common input with every other ORL row. MNIST uses the original three-seed references. Best paper configurations and our repeated means have different reporting conventions. Paper values: Xu et al., IEEE TCSVT (2008), Tables I/II/IV, doi:10.1109/TCSVT.2007.903317.

FERET: three training sizes, PCA and CSA controls

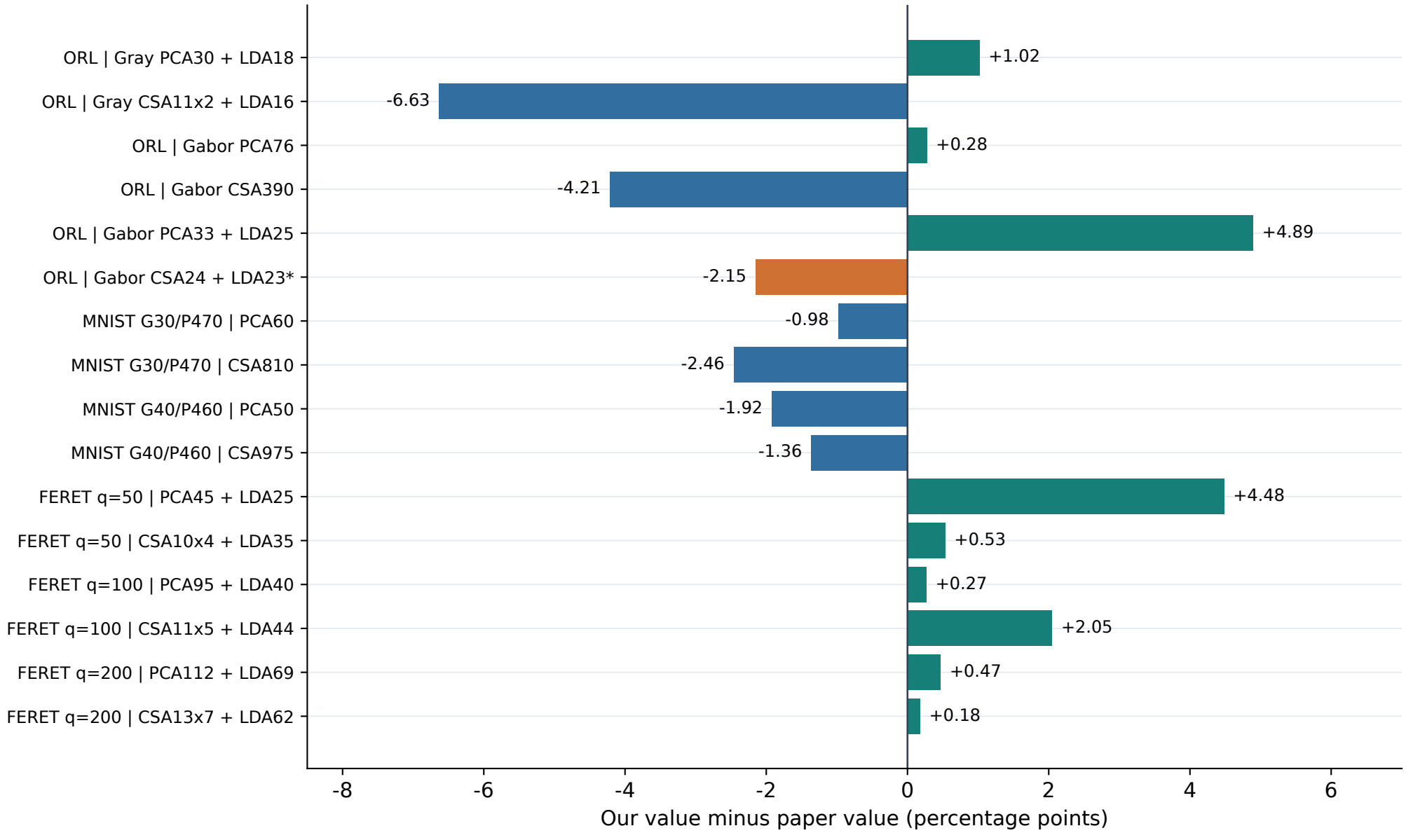
Official grayscale Fa/Fb pairs; first q identities learn the representation; remaining identities are tested.



These are fixed shared-cosine controls, not validation-selected winners. One fixed identity partition per q; no repeated-identity SD or significance claim. CSA does not beat the reproduced PCA at q=50. Paper values: Xu et al., IEEE TCSVT (2008), Tables I/II/IV, doi:10.1109/TCSVT.2007.903317.

Numerical gaps from the paper across all 16 cases

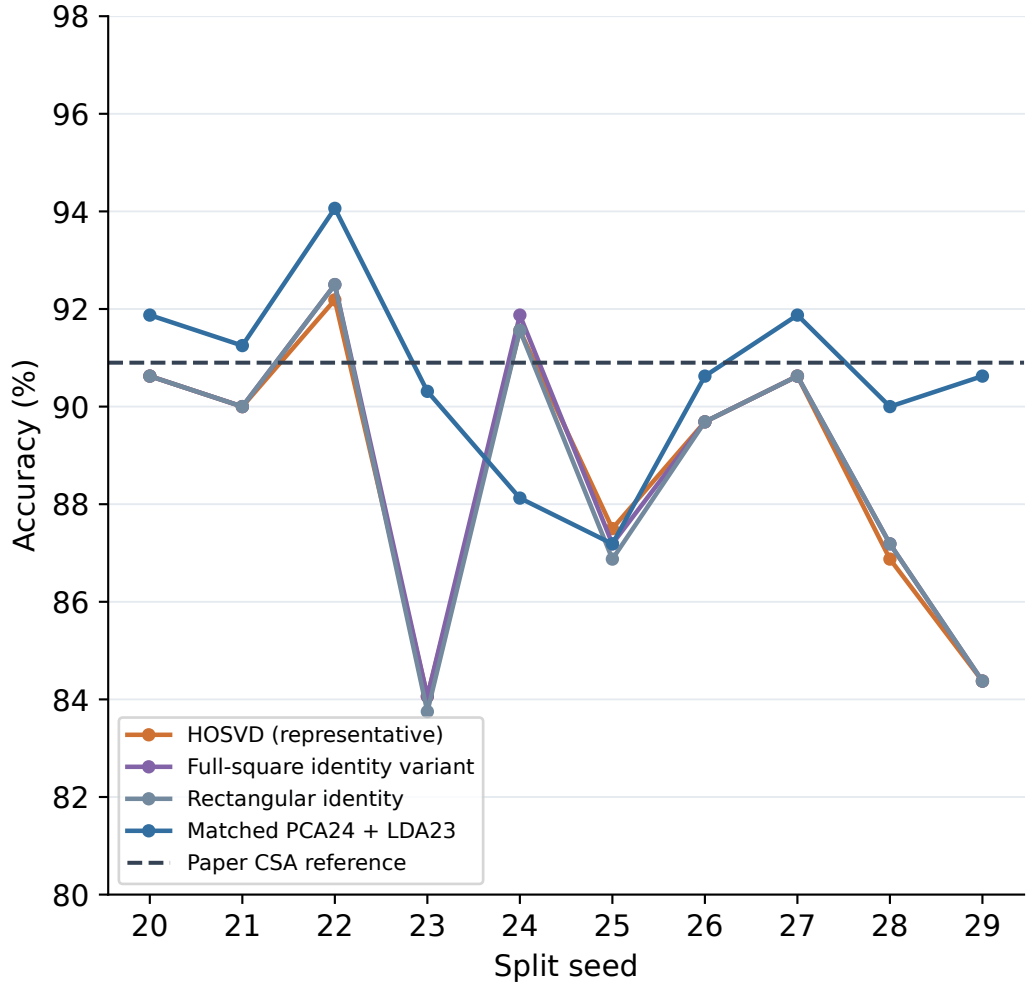
Positive differences mean a higher recorded number; they do not prove a matched-protocol improvement.

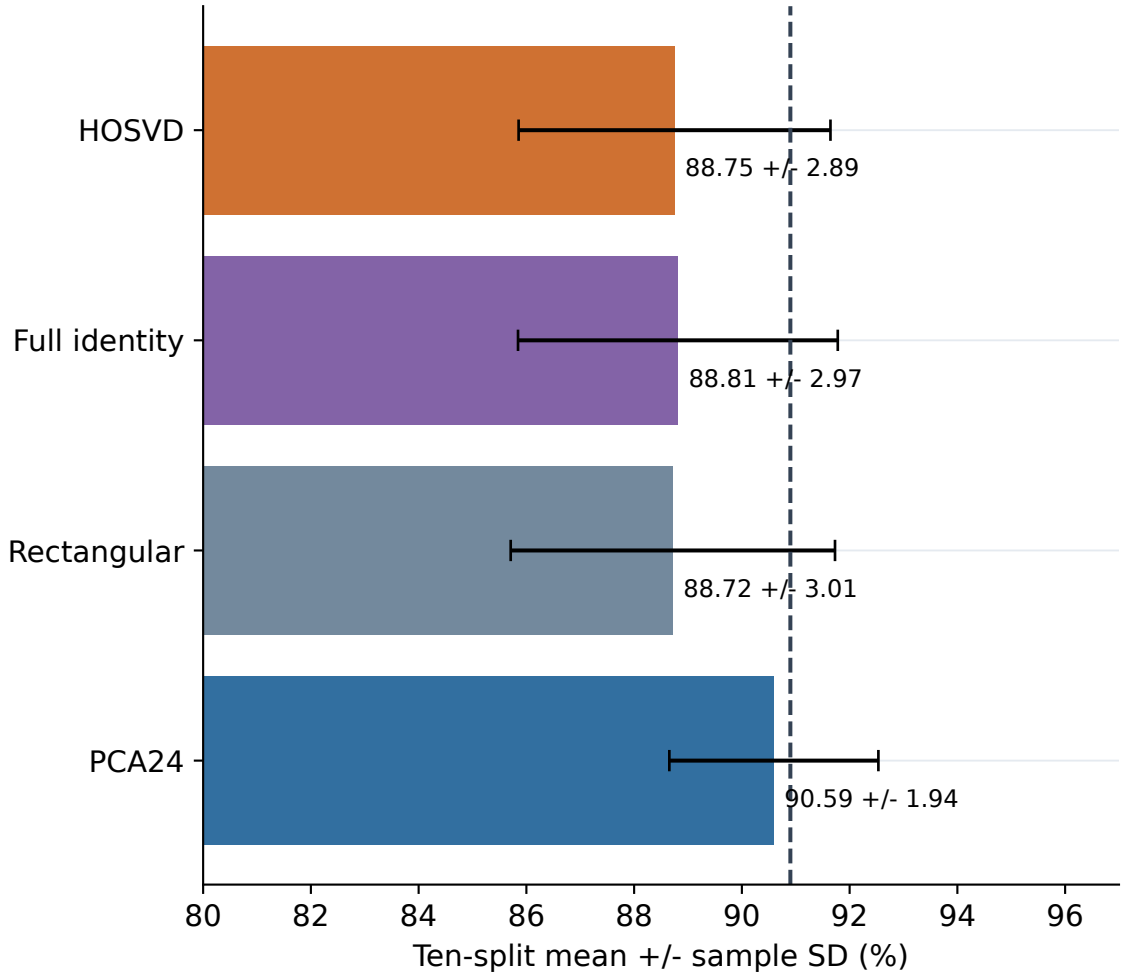


Orange is the HOSVD/context ORL representative. Other values use their preserved shared references. FERET has a fixed identity partition; paper best-result aggregation is not recoverable. Paper values: Xu et al., IEEE TCSVT (2008), Tables I/II/IV, doi:10.1109/TCSVT.2007.903317.

ORL recovery: every one of the ten observed splits

Same input, G2/P8, output 24->23; centered-cosine NN. No split is discarded.

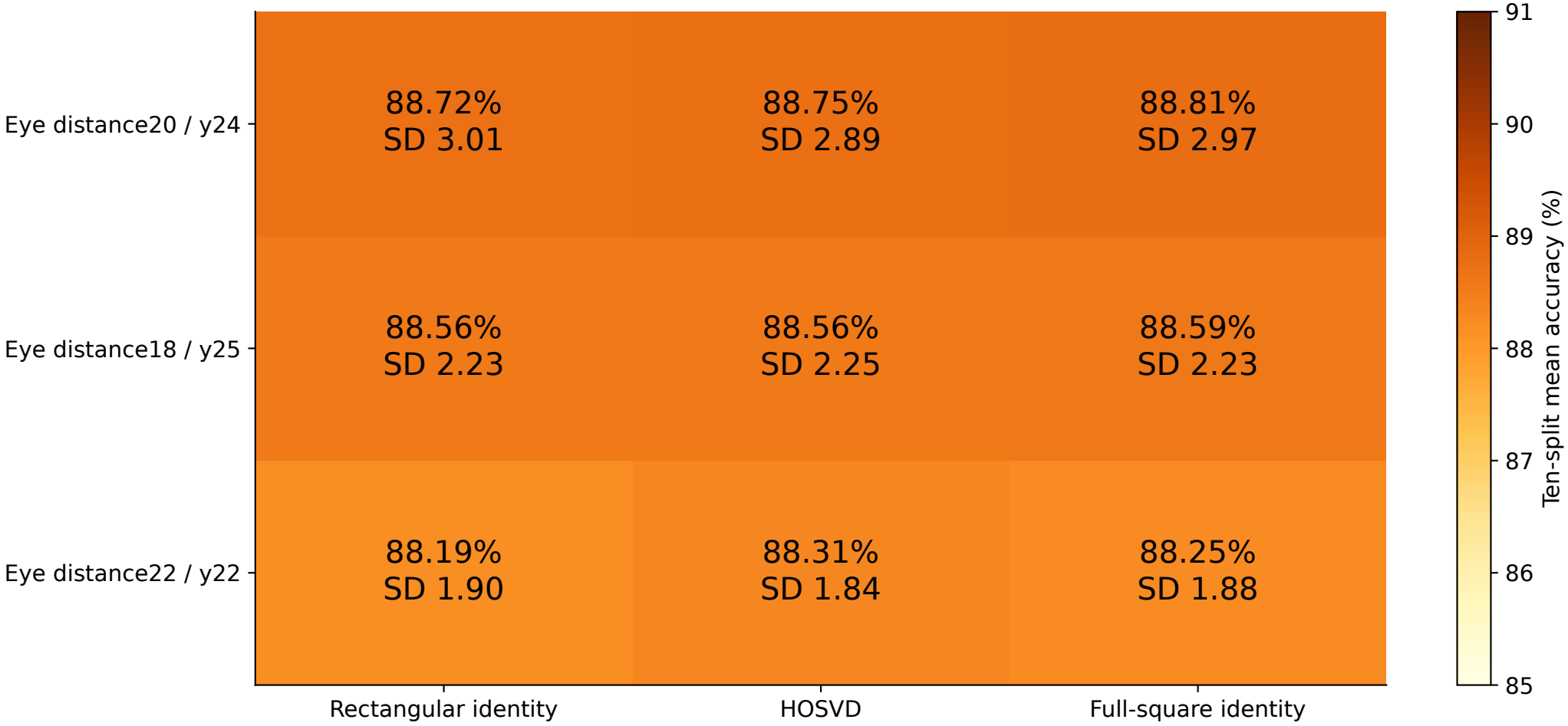




Current CSA representative: HOSVD/context 88.75 +/- 2.89%. Full identity 88.81 +/- 2.97% is a separate start-shape variant. The dashed line refers to CSA, not an original PCA24 result. Observed-split candidate comparisons are exploratory.

ORL: geometry and initialization interactions

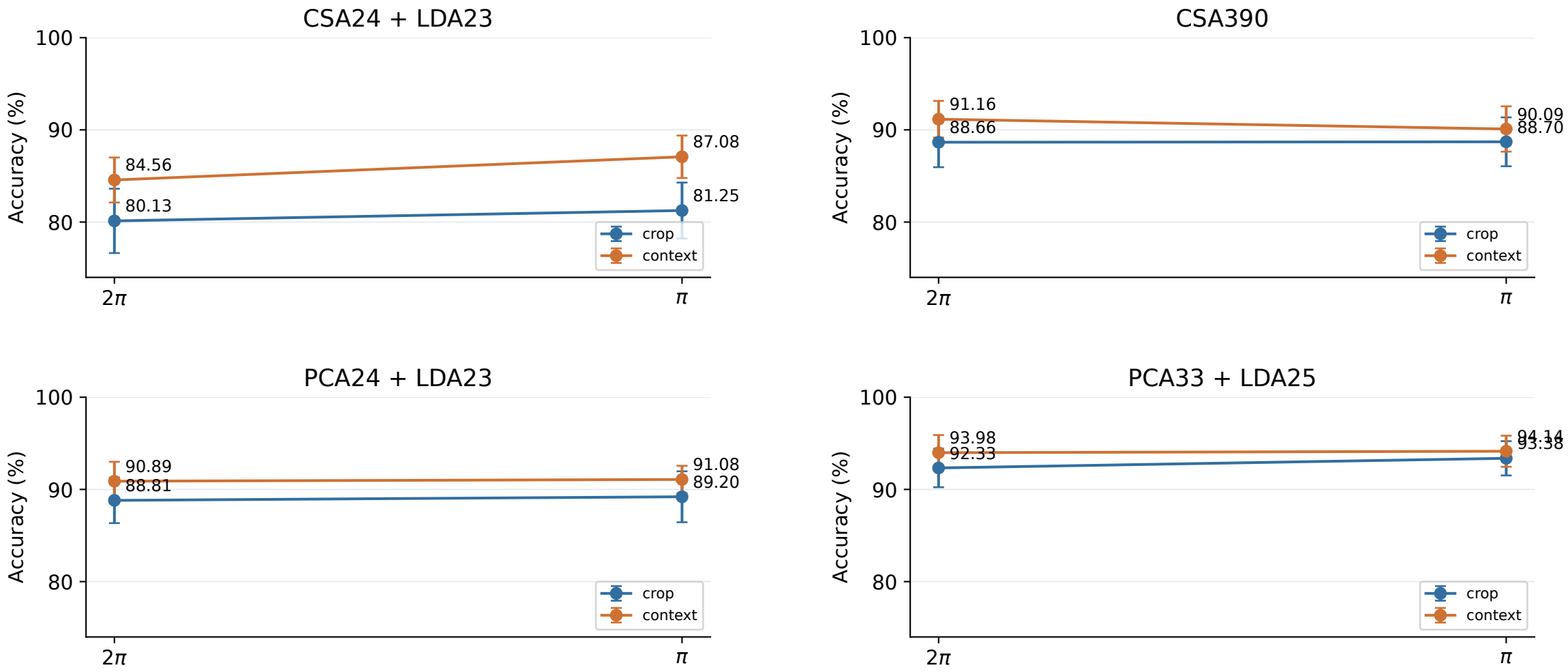
All nine cells: fixed sigma=pi, context, magnitude power1 and the same NN policy.



Geometry bundles change eye distance and height together; their individual effects are not isolated. Full-square initialization begins with larger mode dimensions than the final ranks. None of these means reaches 90.9%.

ORL fixed factorial study: sigma by context

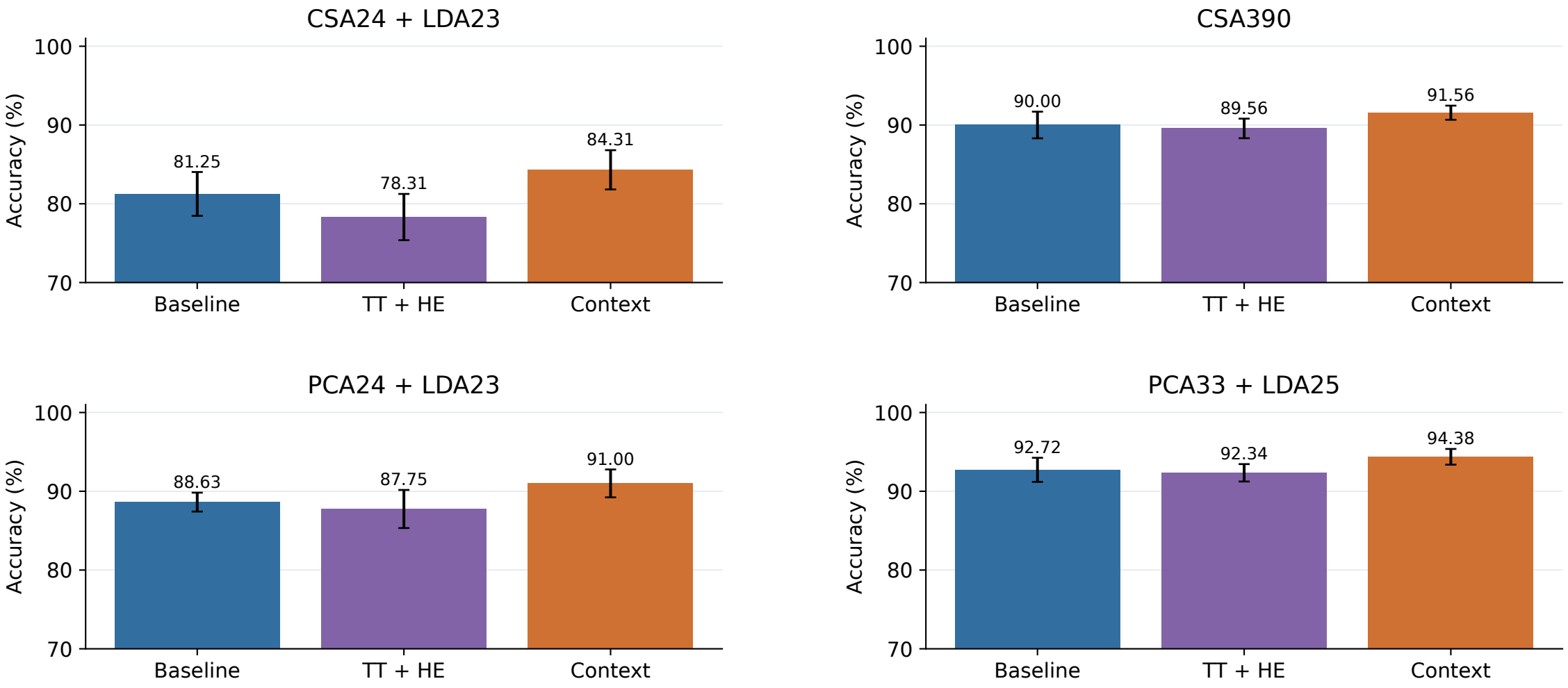
Four methods, twenty predeclared splits, 320 fitted models; fixed settings within each method.



Whiskers are split SD. The effects are descriptive on repeated images. Context improves CSA24, but reducing sigma under context lowers CSA390; the settings are not jointly optimal for every method.

ORL three fixed inputs: improvements and regressions

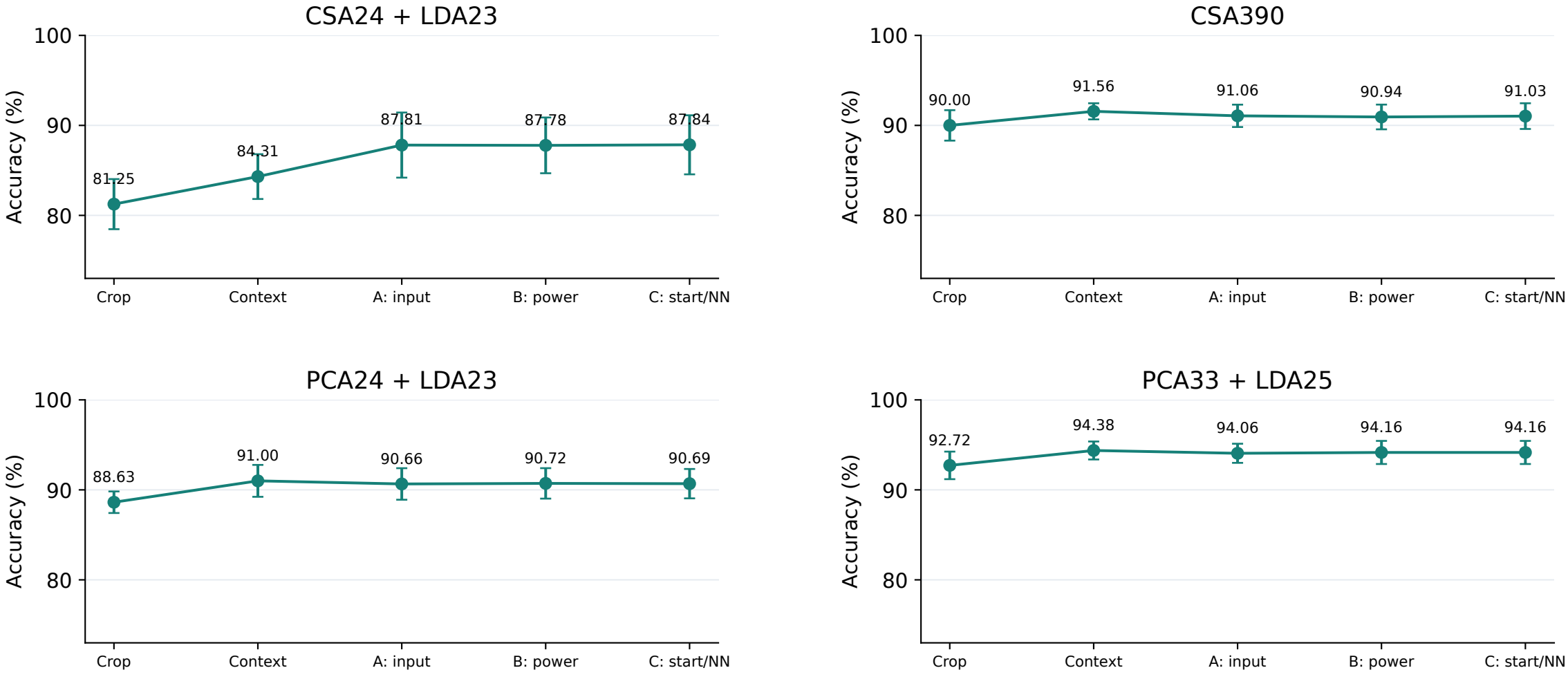
Ten matched splits; baseline, Tan-Triggs plus equalization, and context are tested separately.



120 outer fits; all negative branches retained. Context is a field-of-view extension and TT+HE is a photometric extension. This ten-split batch differs from the twenty-split factorial study.

Gallery-only staged selection: most gain comes early

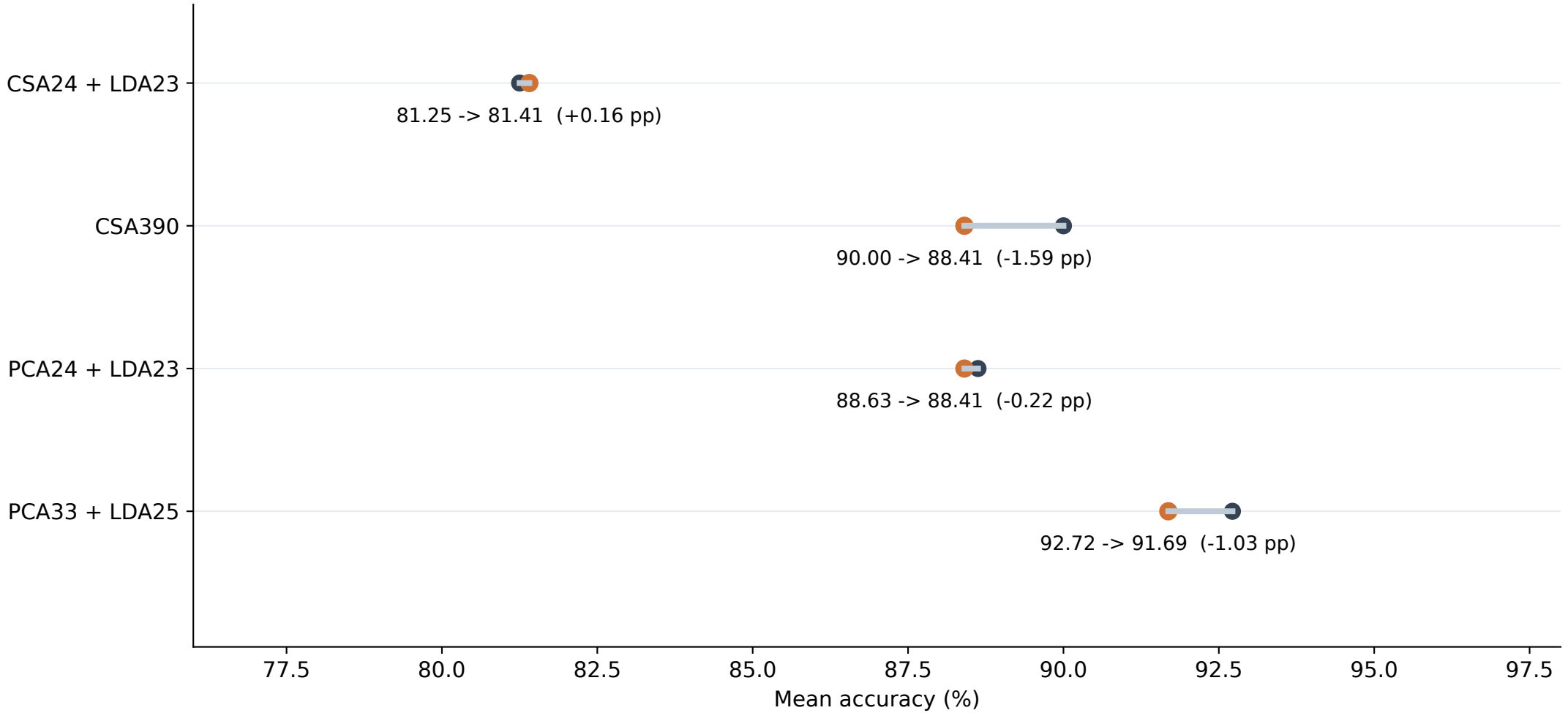
Own-gallery validation; fixed original output dimensions. All ten outer splits are retained.



Inputs include context/power extensions. CSA24 stages A->C add only about 0.03pp, and other controls can decline. Inner validation improvement does not guarantee outer improvement; this is not a blind new dataset.

Crop-only bounded optimization: small gain, negative controls

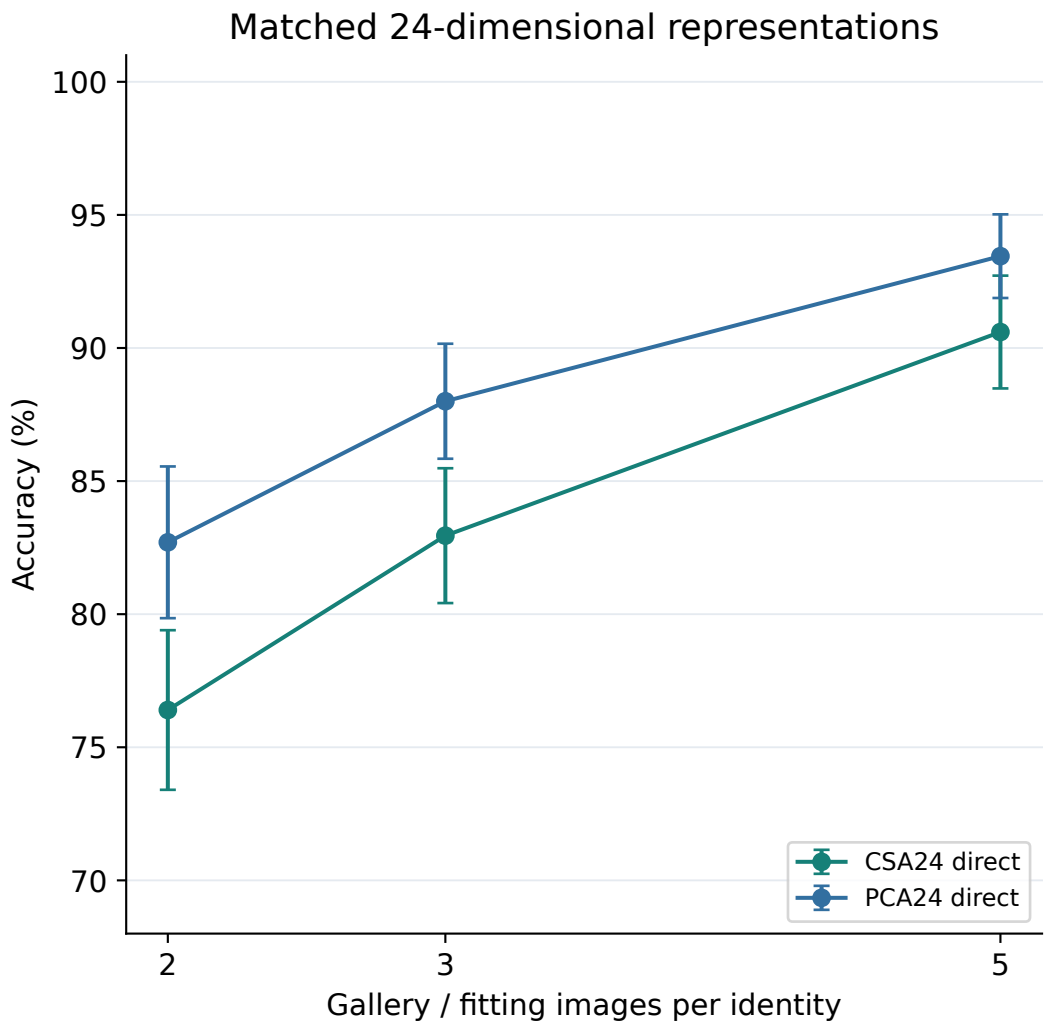
No context, magnitude powers, shrinkage or new ranks; ten paired splits.

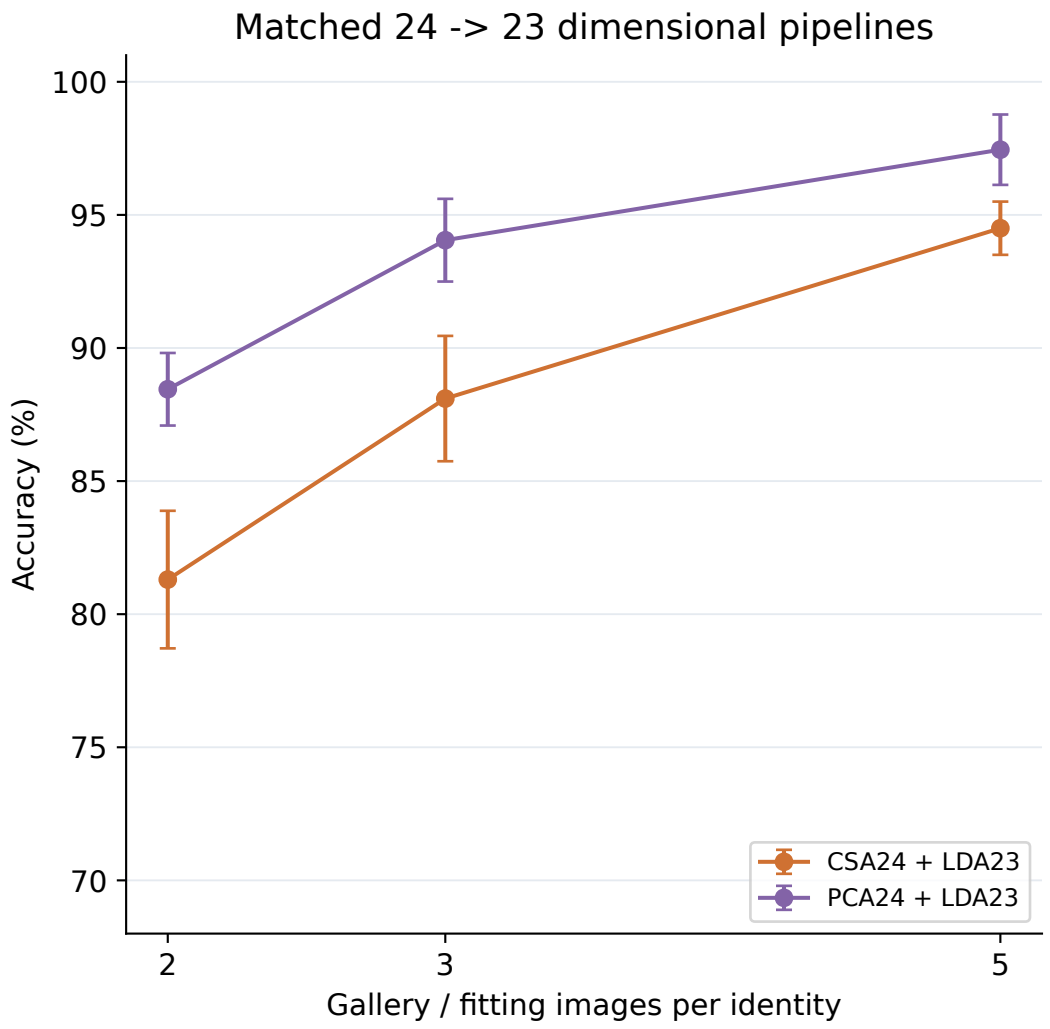


Dark: matched baseline; orange: validation-selected final setting. CSA24 improves only 0.16pp, while CSA390 and both PCA controls decline. Do not substitute this result for the context extension.

ORL sample-size extension with the same five probes

Nested G2 subset G3 subset G5; the held-out P5 images stay fixed within each split.

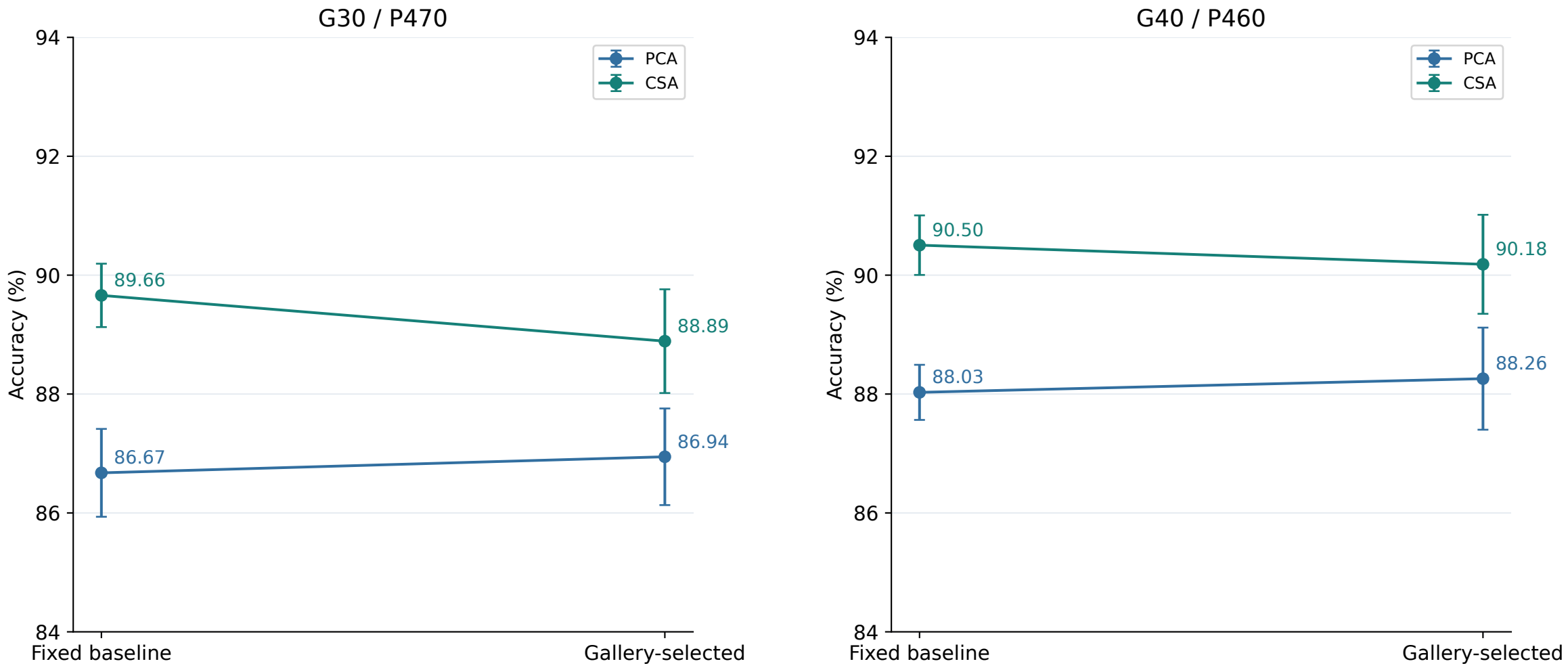




Ten splits; sample SD whiskers. More fitting/gallery images change both representation estimation and NN coverage. G5/P5 is an extension, not the original G2/P8 experiment; no original-paper threshold is drawn.

MNIST bounded selection: preserve the negative result

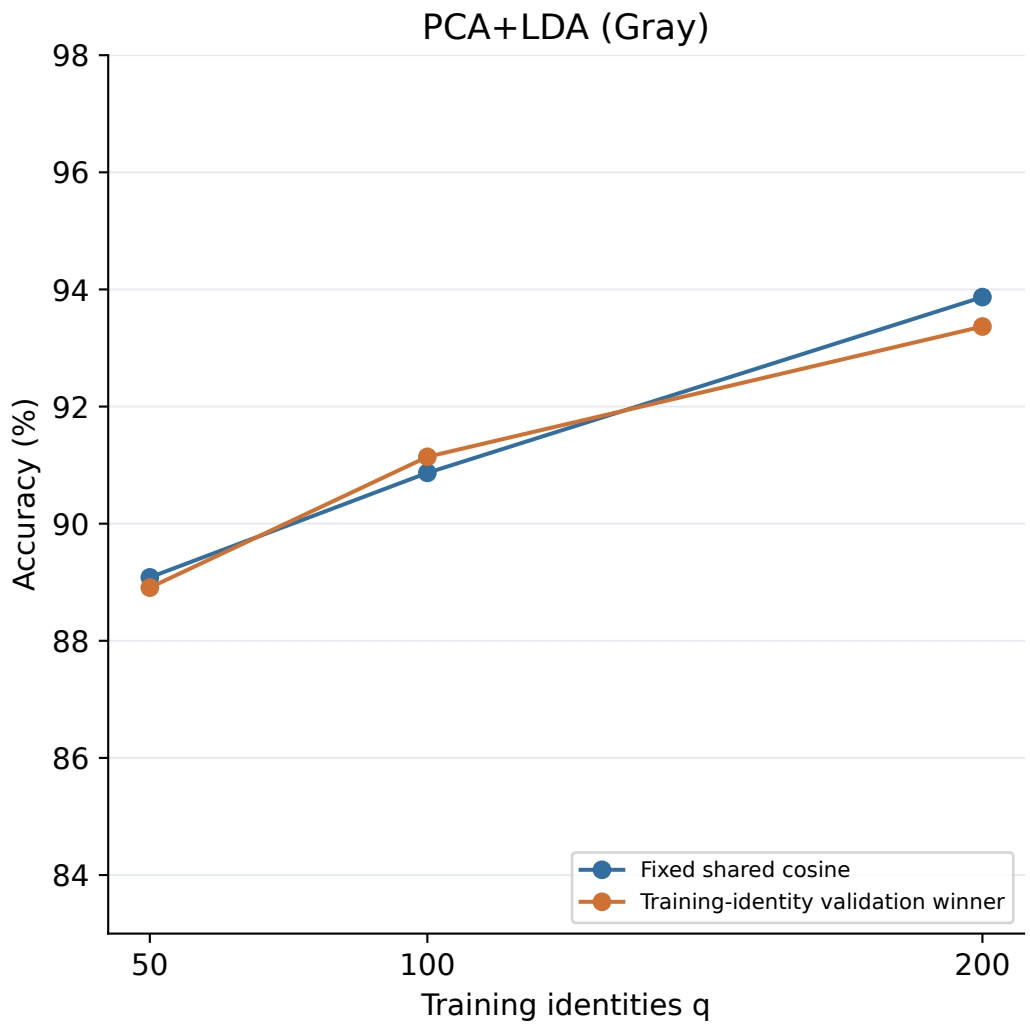
Ten matching splits per gallery size; PCA and CSA use their original output dimensions.

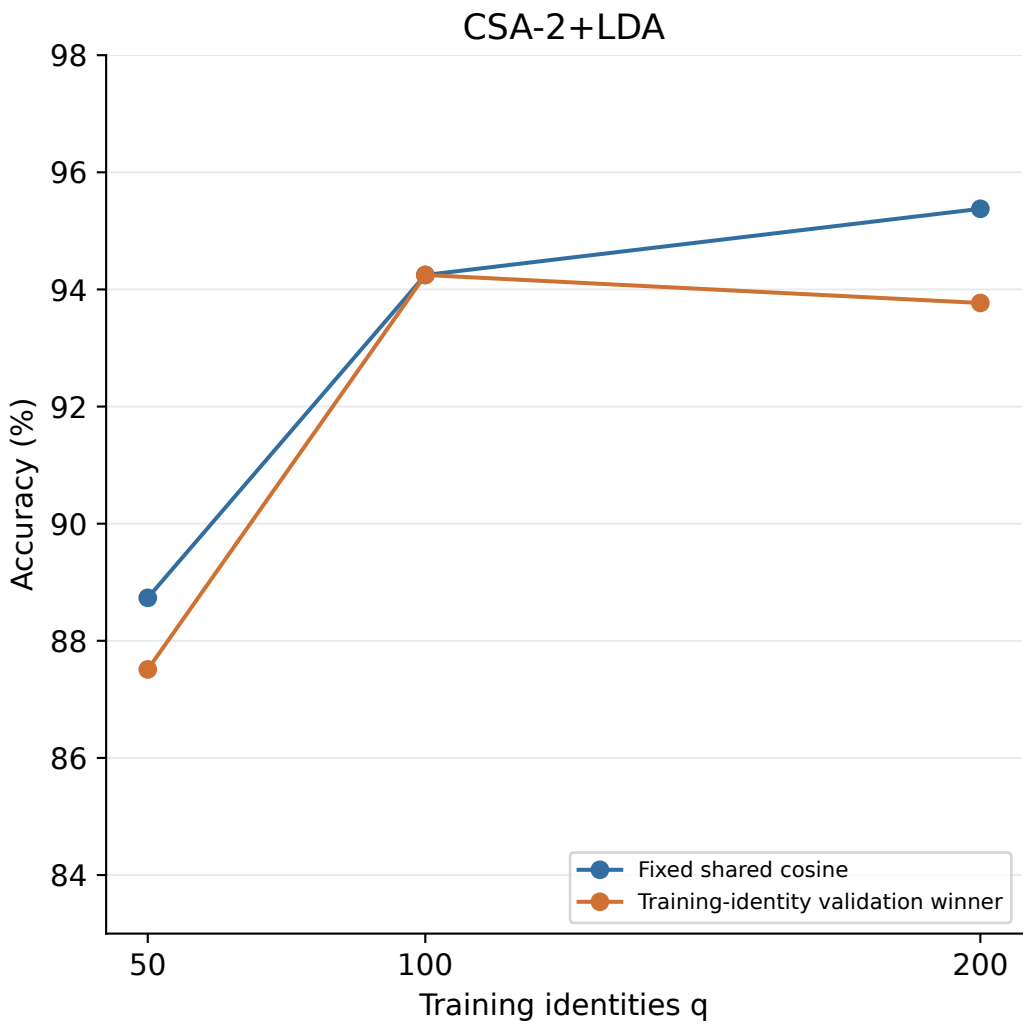


CSA declines by 0.77pp at G30 and 0.32pp at G40. These are ten-seed paired results, not changes from the three-seed main-reference means. All failed stages remain archived.

FERET: validation winners and fixed controls differ

Do not relabel the fixed shared-cosine scores as validation-selected winners.

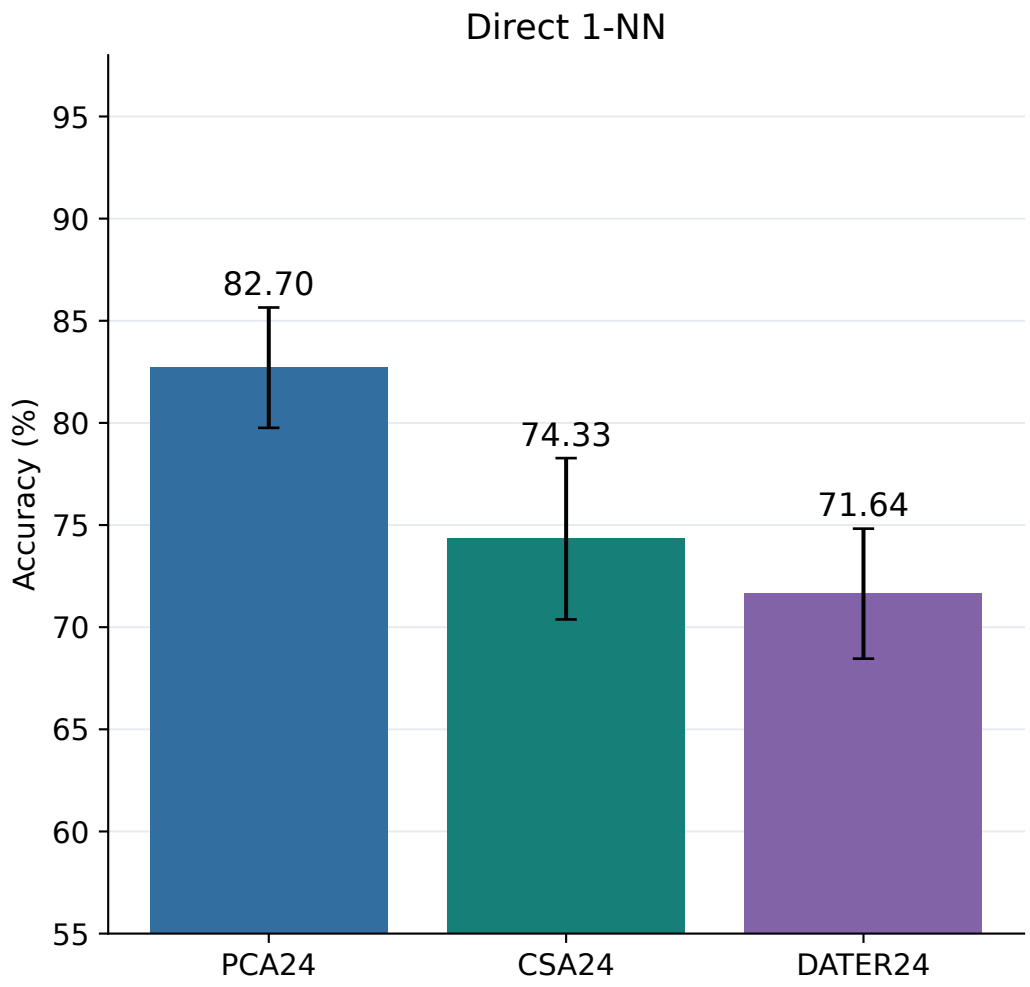


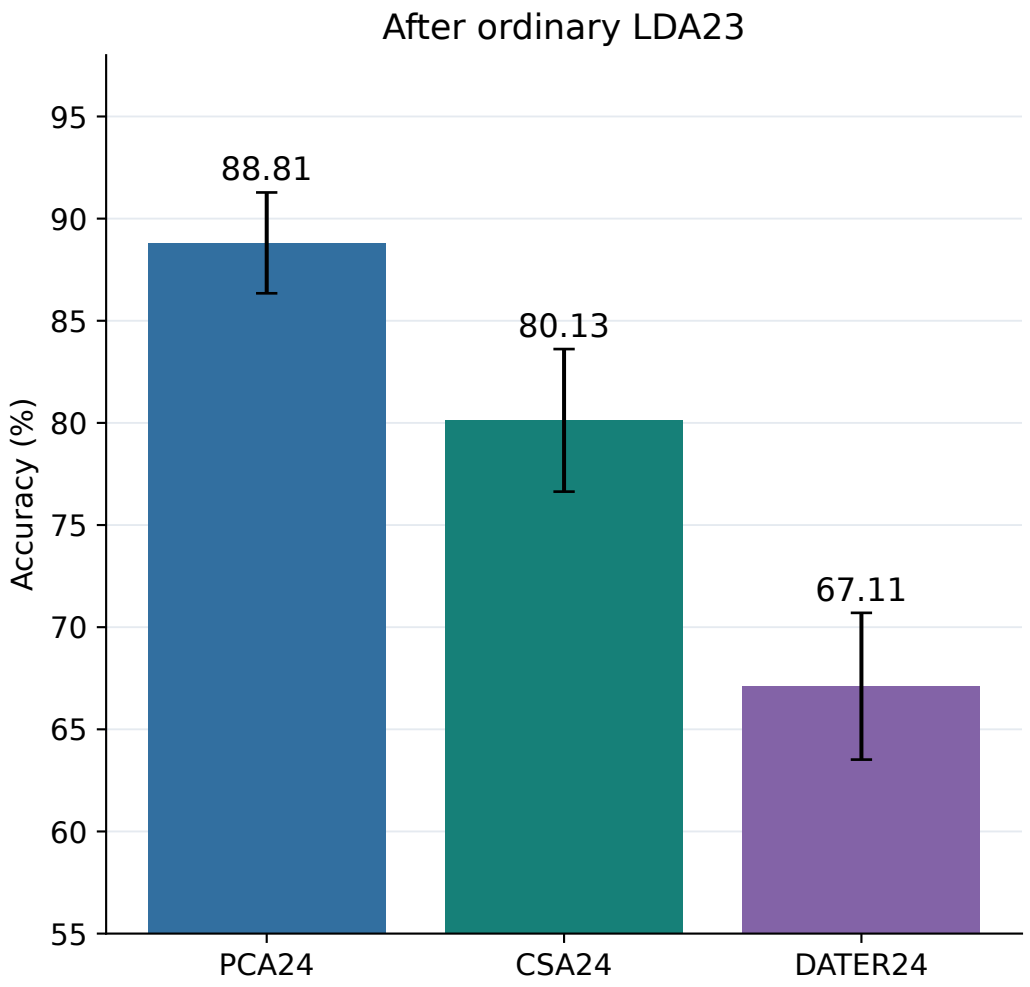


One fixed identity partition per q; no split SD. For q200 CSA, the fixed-cosine 95.38% is higher than the actual validation-selected 93.77%; both are retained.

DATER supervised tensor extension: a negative result

Matched 2pi-crop input, twenty splits, 24D representation; ordinary LDA output23.

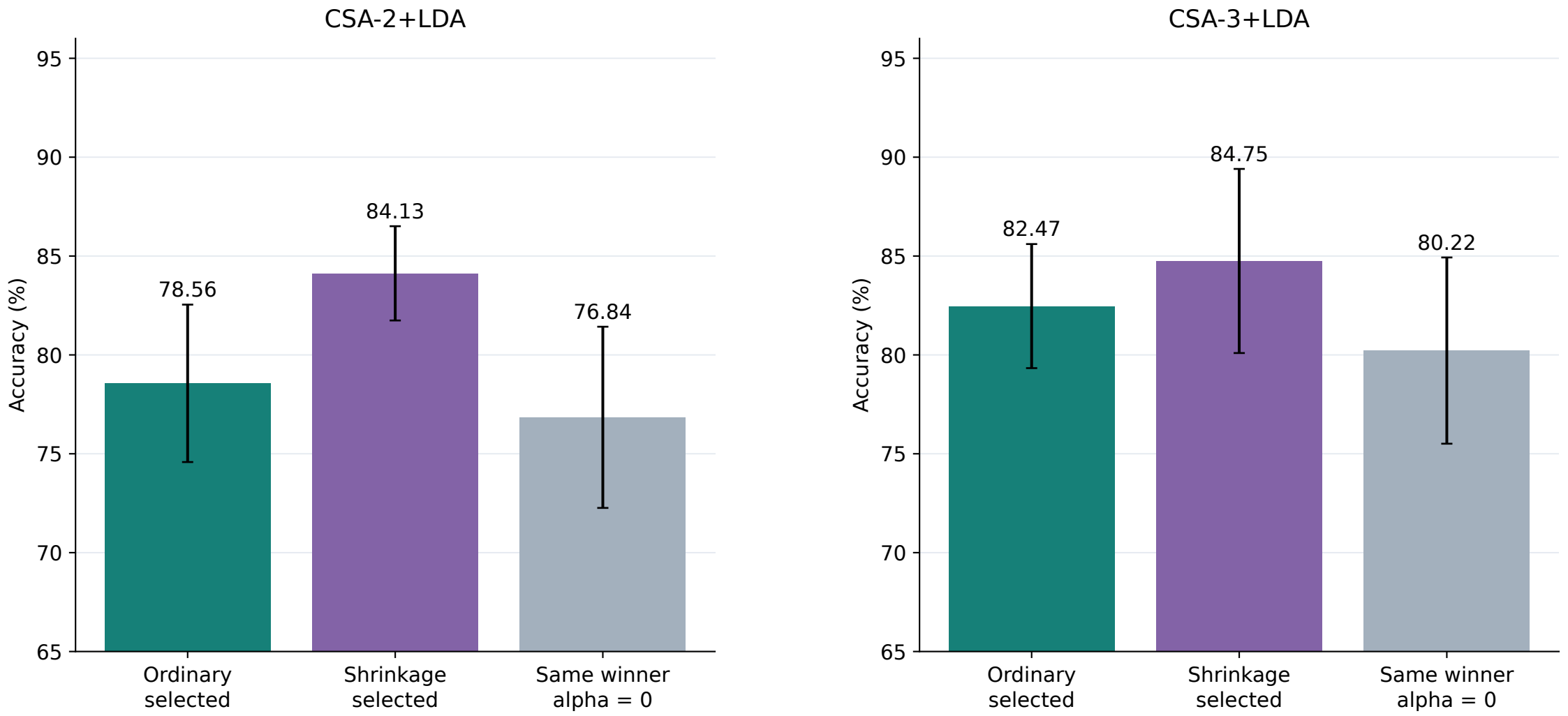




DATER: 20 new models; CSA/PCA controls reused. 19/20 DATER runs hit the20-iteration cap. Strict free-trajectory failure and numerical sensitivity are recorded; this result does not establish that DATER is generally ineffective.

Shrinkage ablation: compare both selected and matched variants

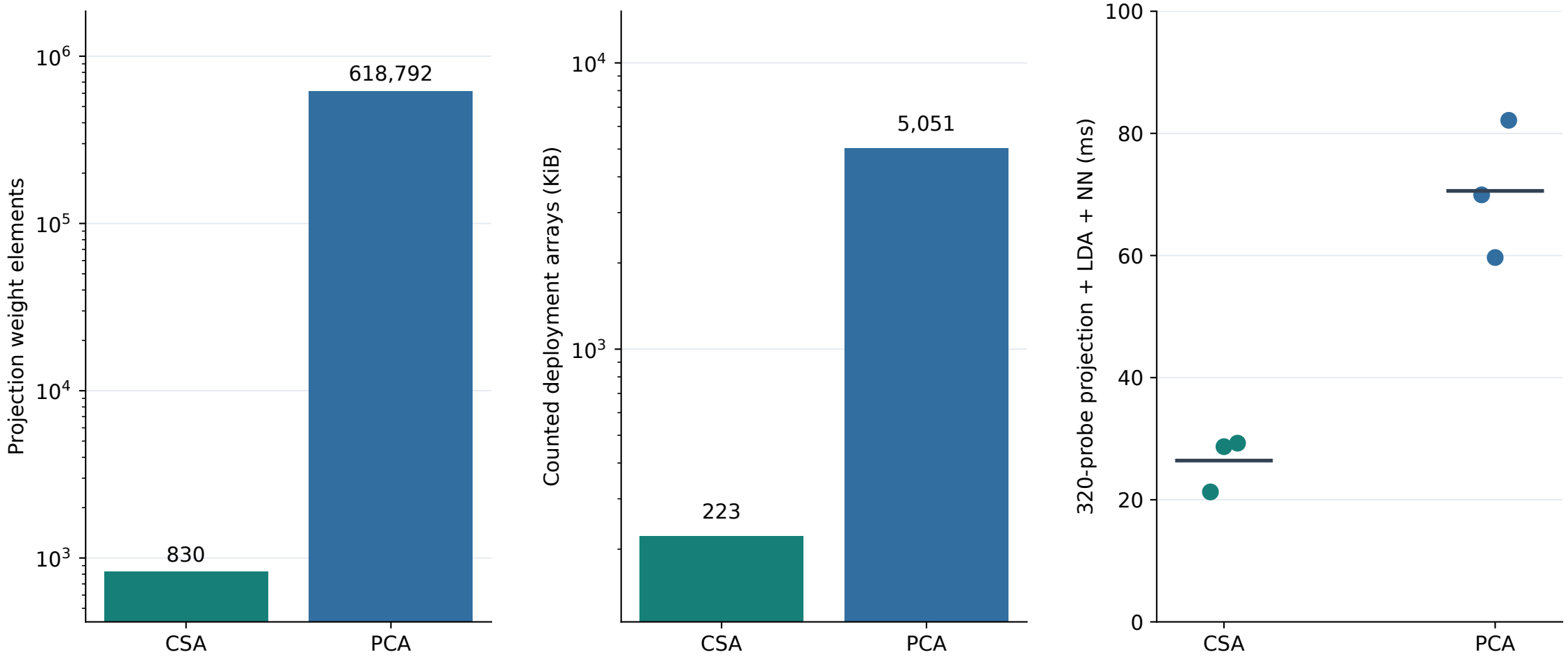
Ordinary and shrinkage pipelines are selected separately; alpha0 removes shrinkage from the selected winner.



Ten splits; SD whiskers. Ranks/input settings may differ between independently selected ordinary and shrinkage pipelines. Only the same-winner alpha0 comparison isolates removal of regularization in that selected configuration.

Representation cost: small weights, explicitly scoped timing

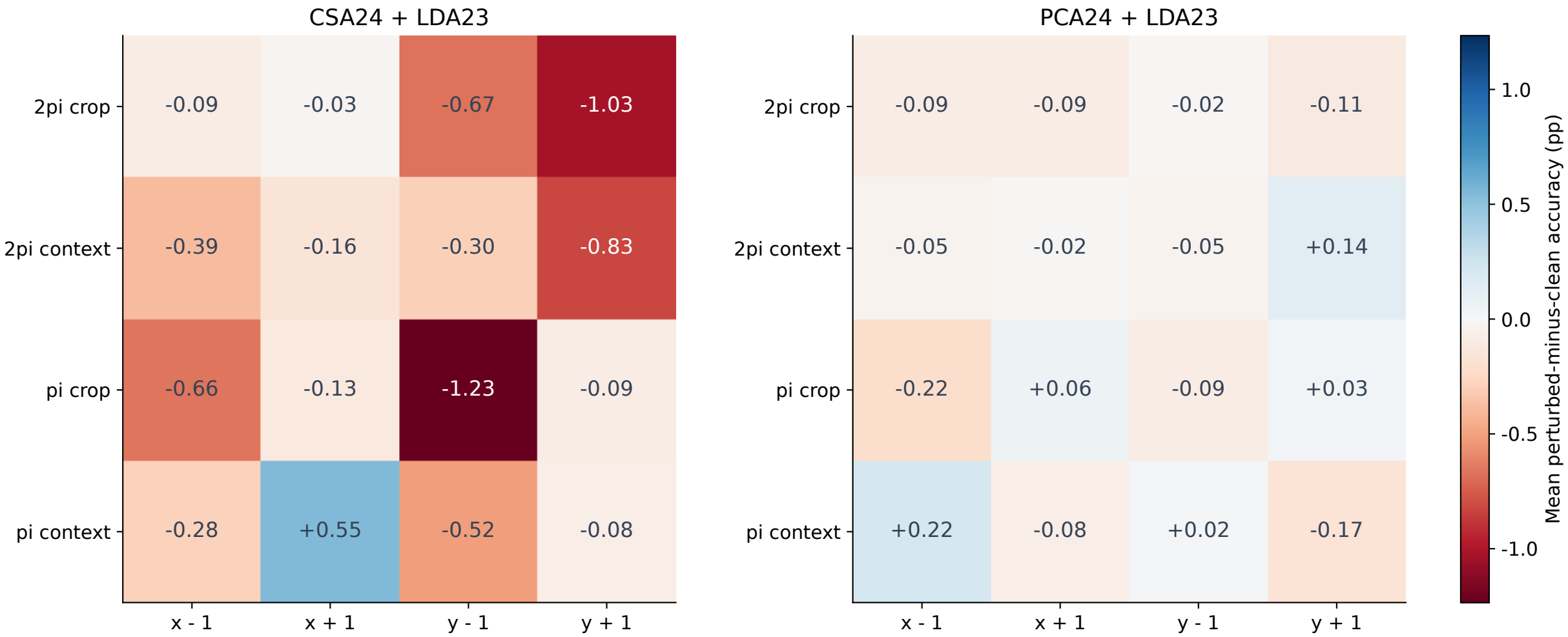
Matched CSA24/PCA24 + LDA23; local Core Ultra9 275HX, two BLAS threads.



Weight counts830 vs618792; arrays228272 vs5171968 bytes. Timing points are per-seed medians, not an uncertainty estimate. Excludes alignment, Gabor, training and I/O; counted arrays are not peak system memory.

Probe-eye perturbations: local robustness with no refitting

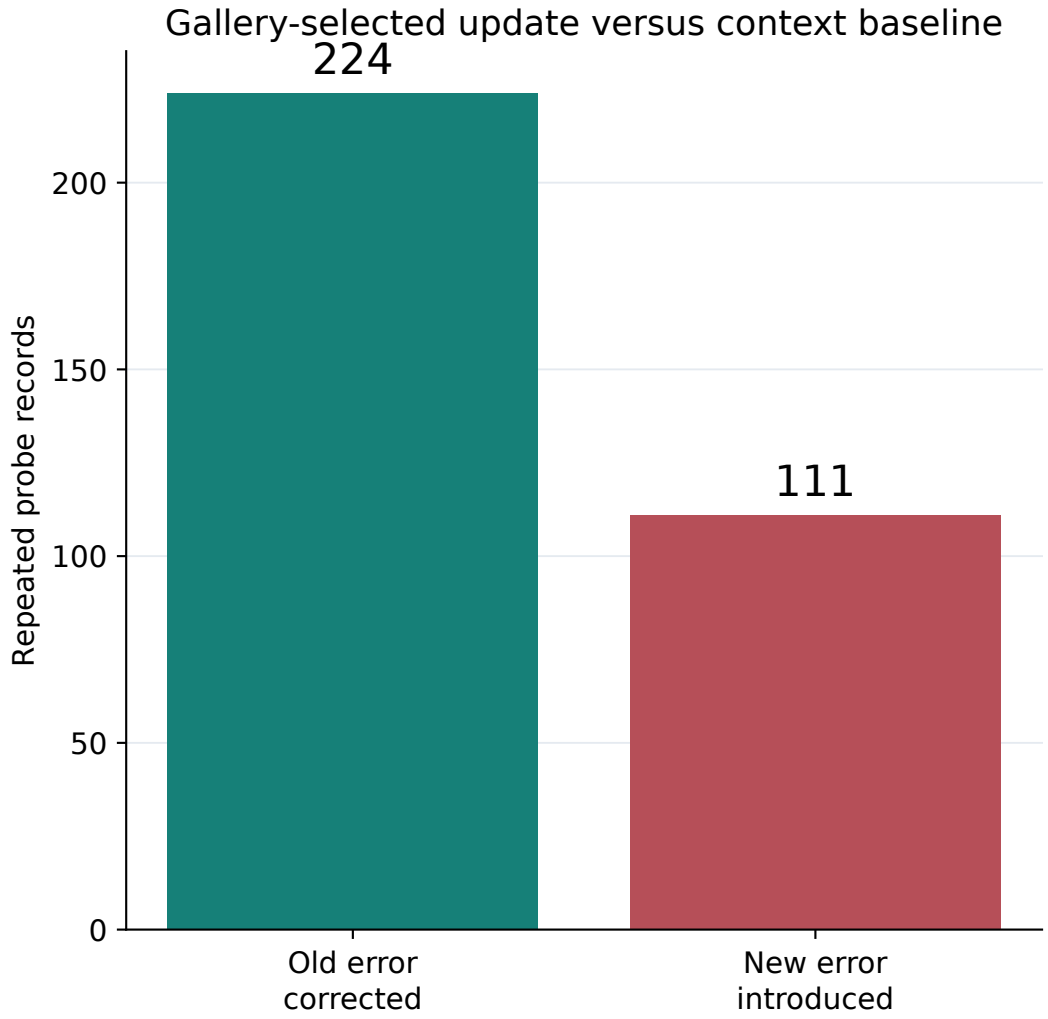
Both eye coordinates translated together by one source-image pixel; models/gallery remain fixed.

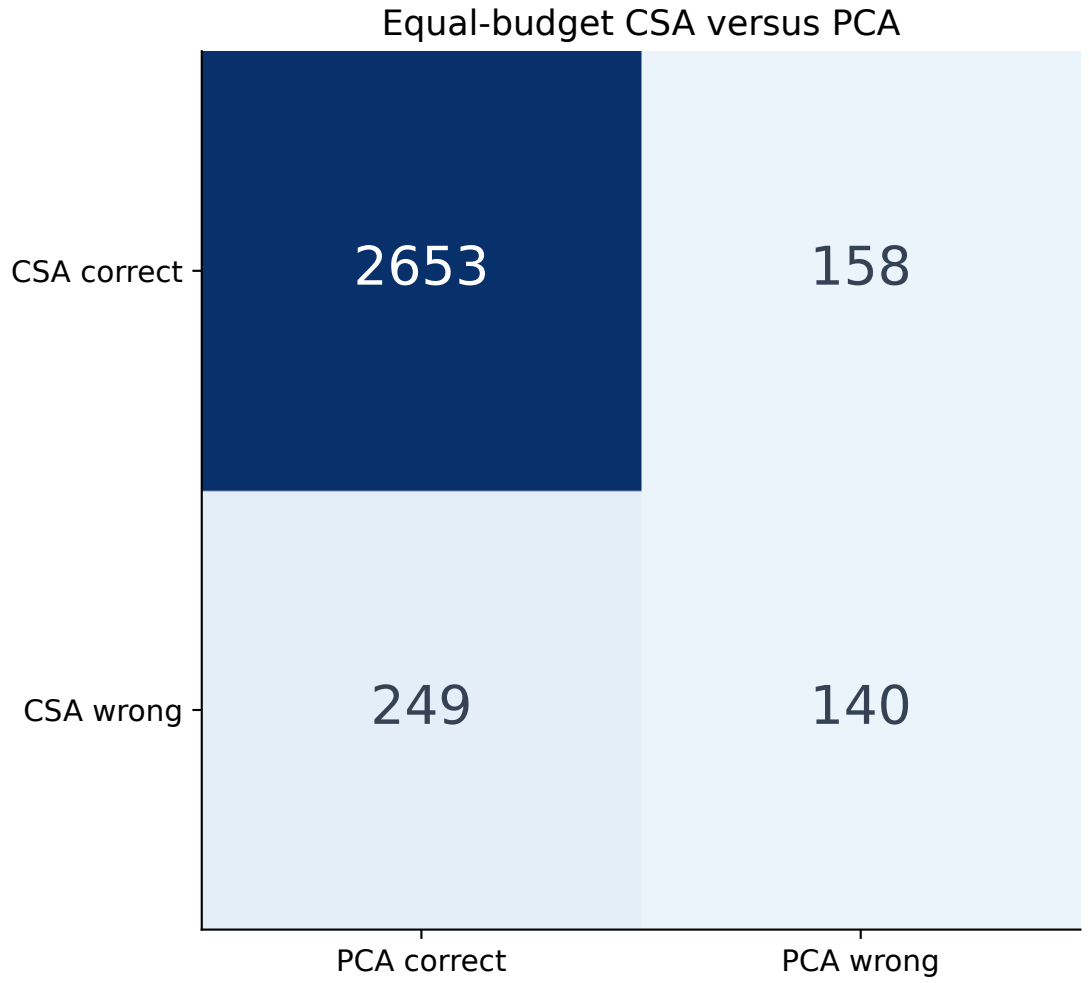


Twenty overlapping splits. The full robustness batch has1280 scores and409600 repeated predictions, zero new fits. This does not cover single-eye errors, rotation or large offsets; no universal robustness claim.

Error analysis: corrections coexist with new failures

Different panels answer different comparisons; counts are repeated probe records, not independent photos.

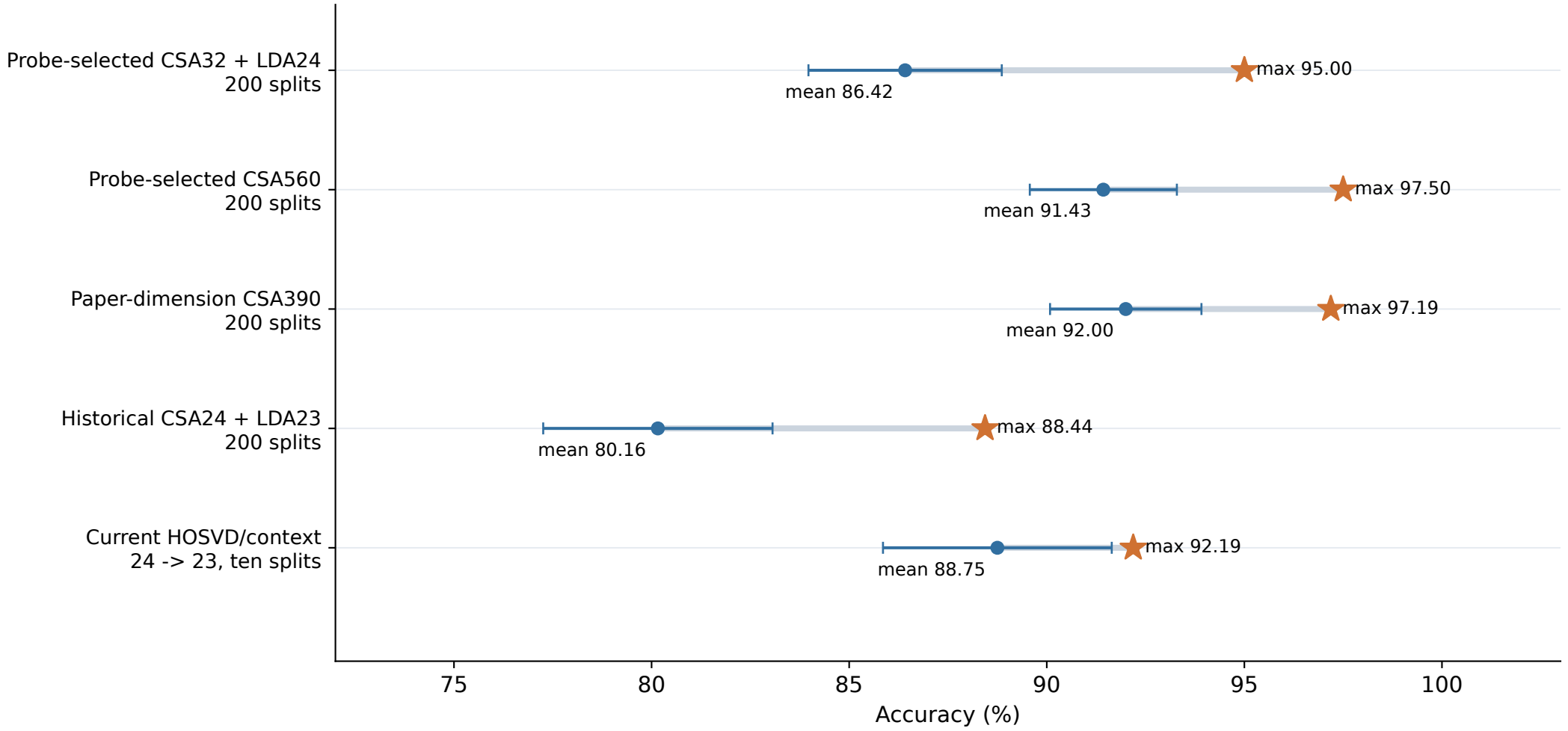




3200 repeated probe records cover400 distinct photos. Input update:224 corrected and111 introduced errors. Same-budget comparison: CSA-only158, PCA-only249. Examples do not establish one causal explanation.

Selected configurations: mean versus observed maximum

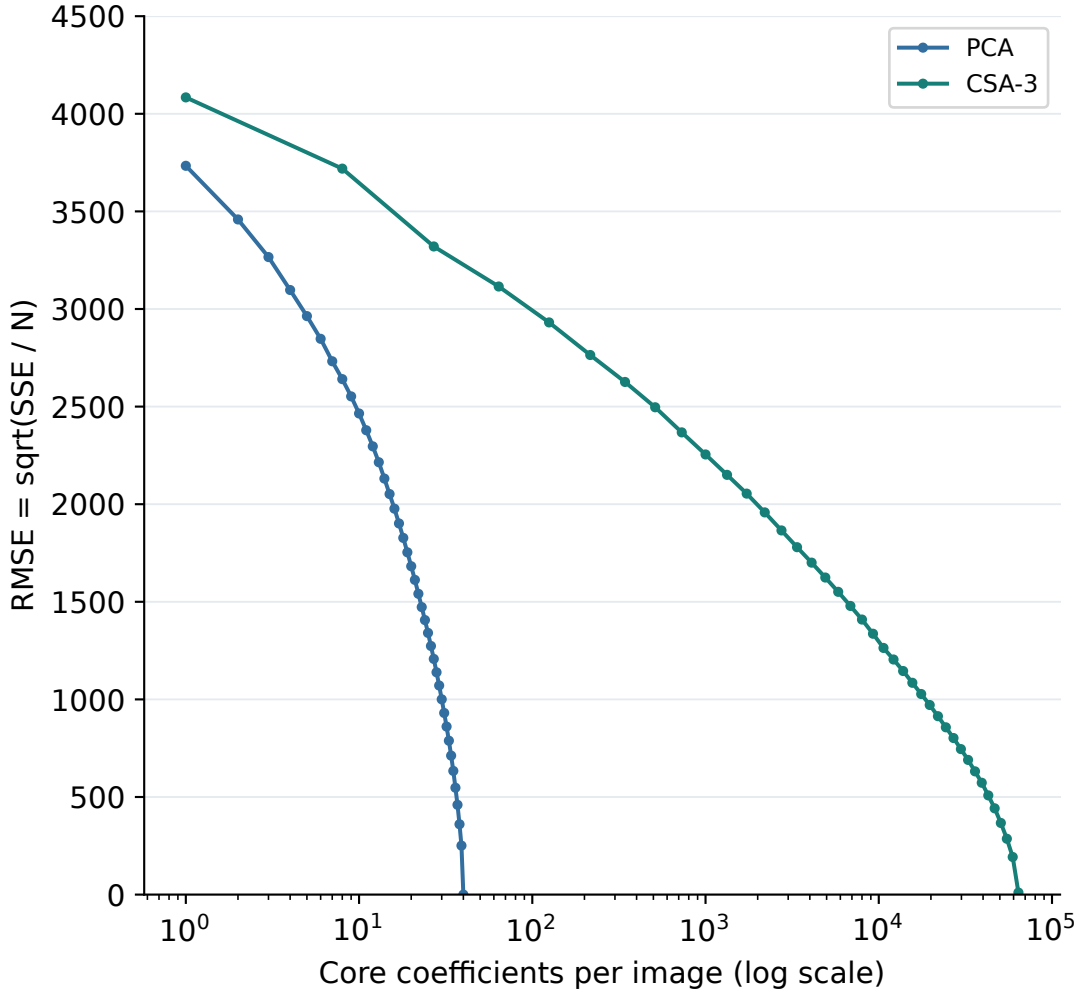
A high single run and the mean of its configuration are different quantities.

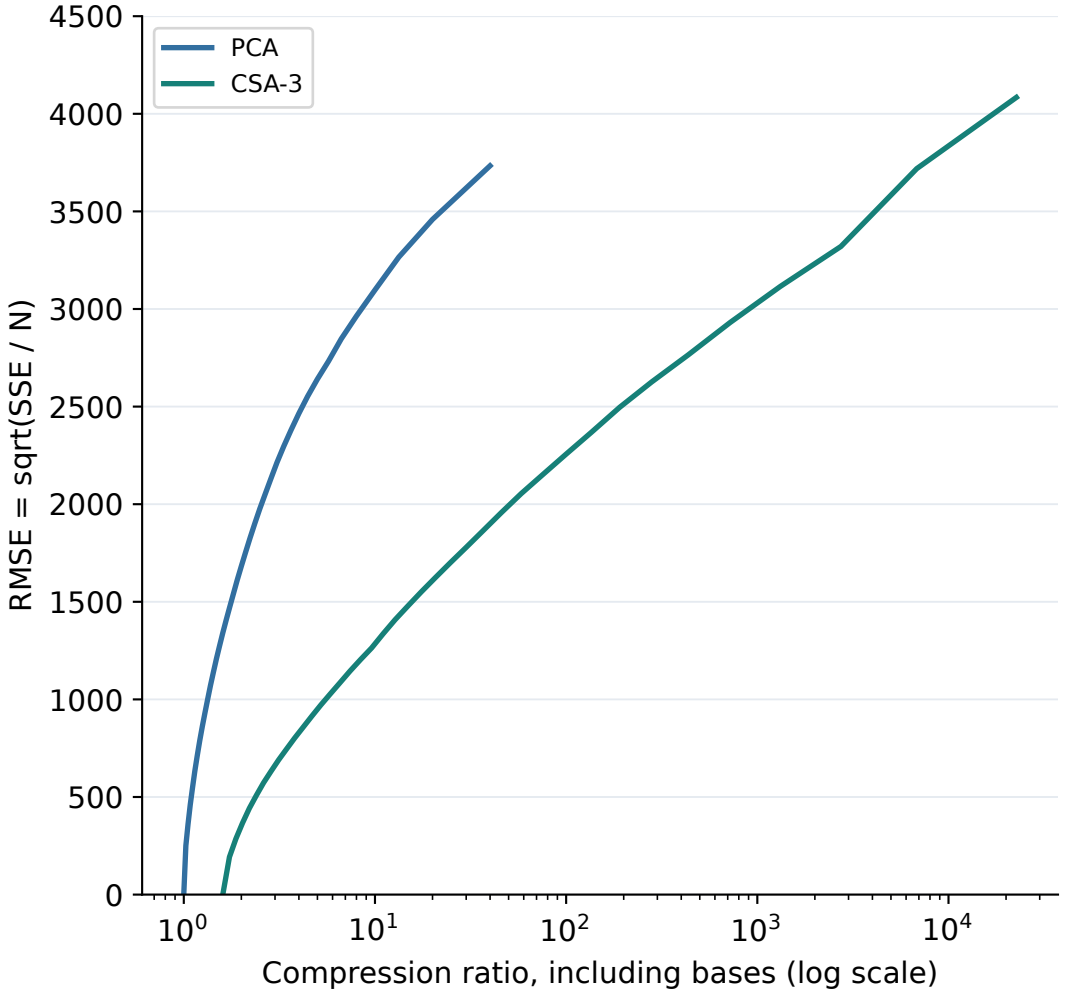


Historical winning configurations were selected using probe scores across configurations and splits. Their means are also post-selection summaries. Different dimensions, inputs and split counts prevent a common-protocol ranking; maxima are not stable test estimates.

ORL tensor reconstruction: error and representation storage

Forty fitting images; 56x46x40 Gabor tensors; original RMSE definition sqrt(SSE/N).

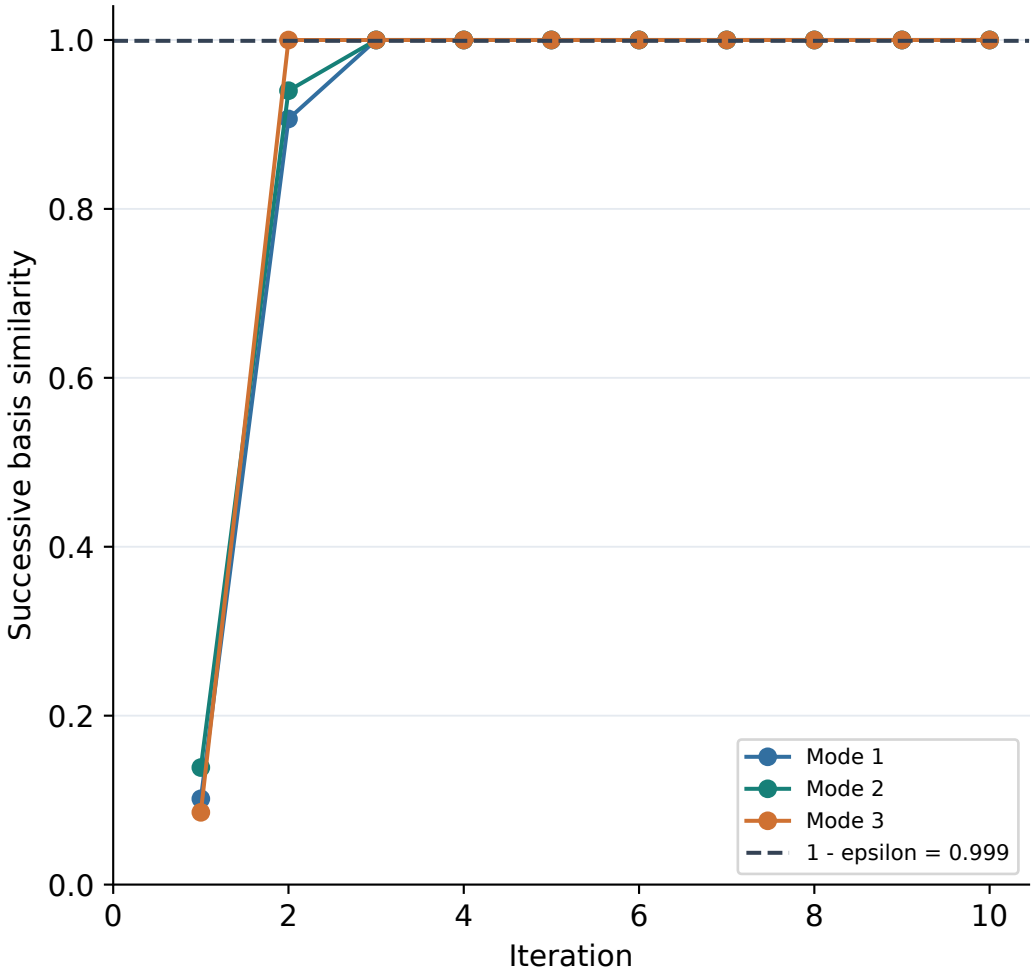


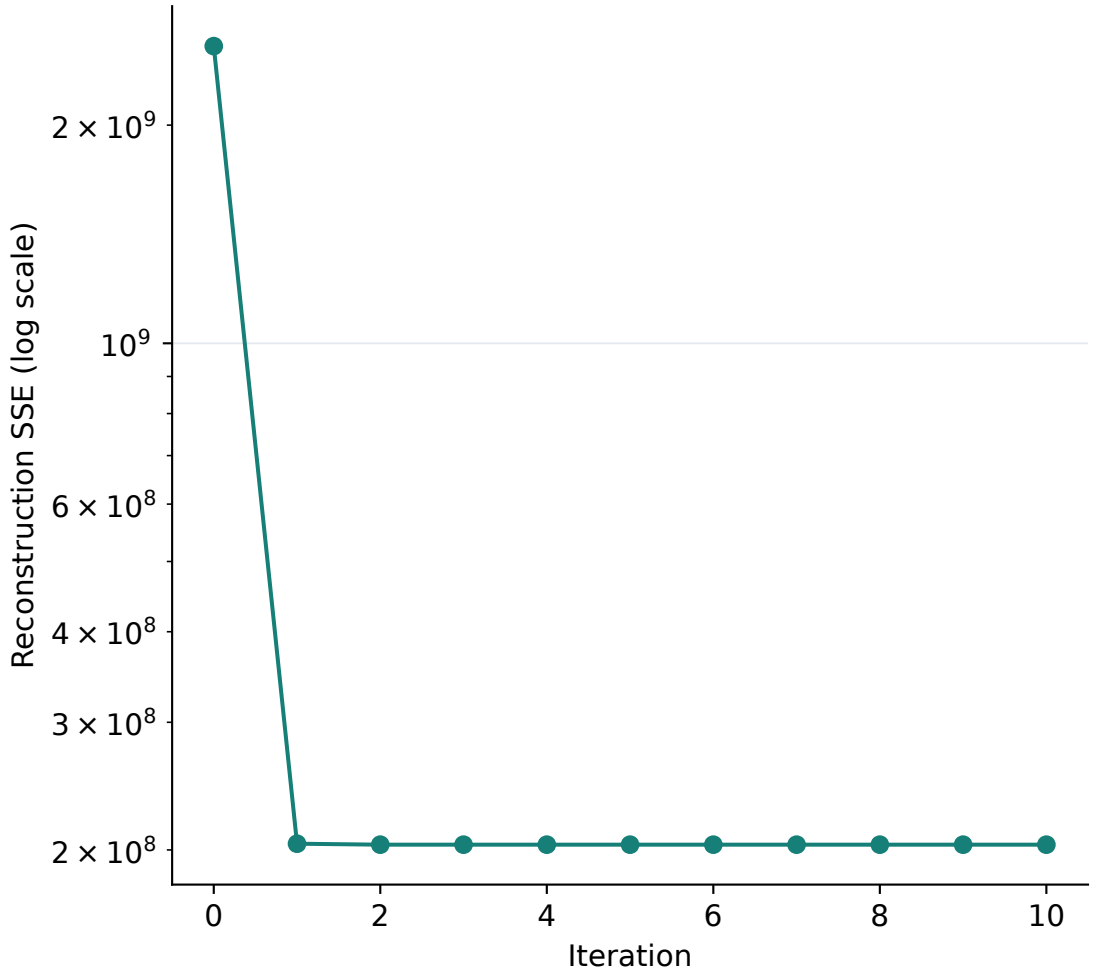


Historical Gabor/alignment version; not the new HOSVD/context recognition input. Learning-sample tensor reconstruction, not Gabor inversion or unseen-face generalization. Original absolute Gabor scale/pointwise curve values are unavailable; compare qualitative trends.

CSA convergence: rapid stabilization in the recorded input

One image per identity; ten diagnostic iterations with early stopping disabled for plotting.

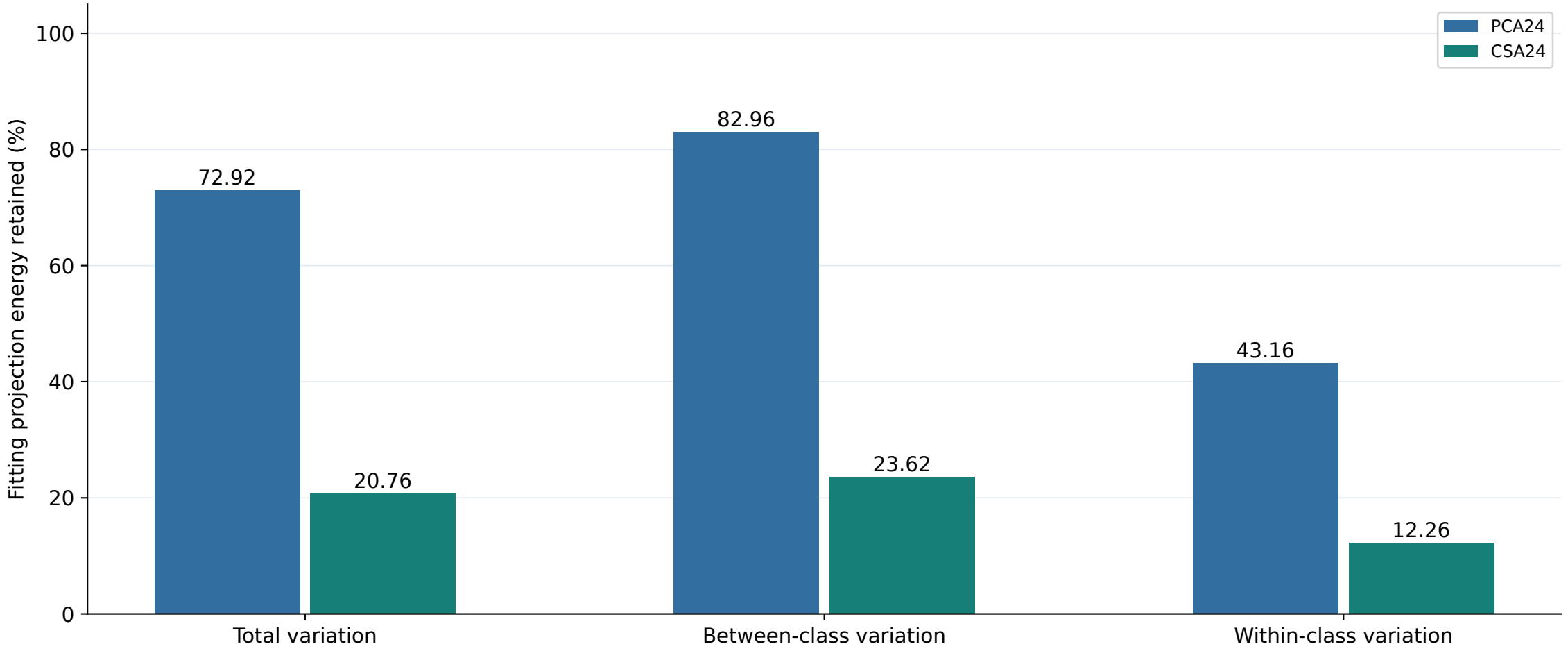




This diagnostic does not replace the recognition limitTmax3. Rapid SSE/basis stabilization does not prove a global optimum or optimal classification. Exact paper ordinates and author input scaling are not recovered.

Equal dimension does not imply equal retained variation

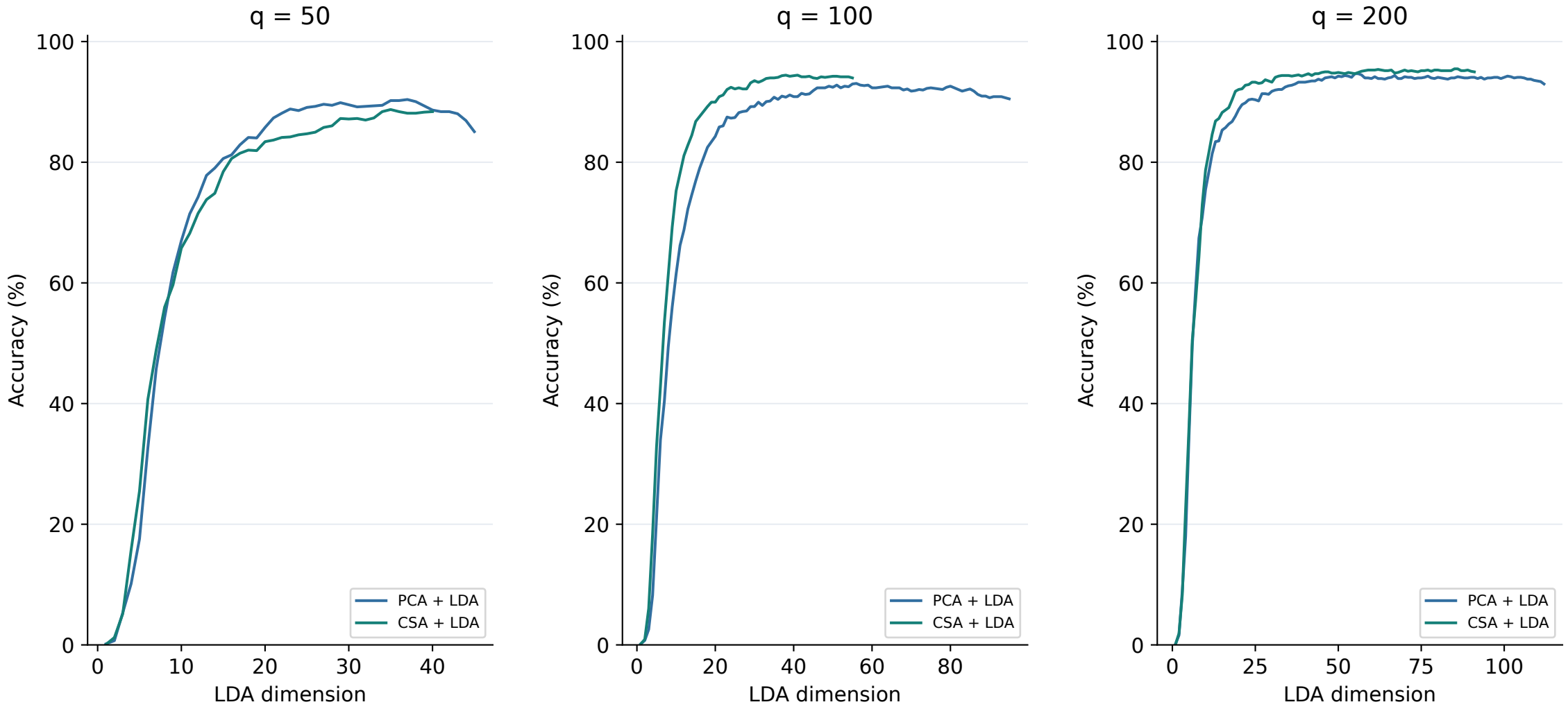
Historical matched-input Gabor24-dimensional representations; fitting-data diagnostics.



These are projected-energy ratios, not percentages of identity information or accuracy. PCA and CSA have different structural constraints and centering conventions. This diagnostic alone does not identify the cause of the recognition gap.

FERET main experiment: recognition versus LDA dimension

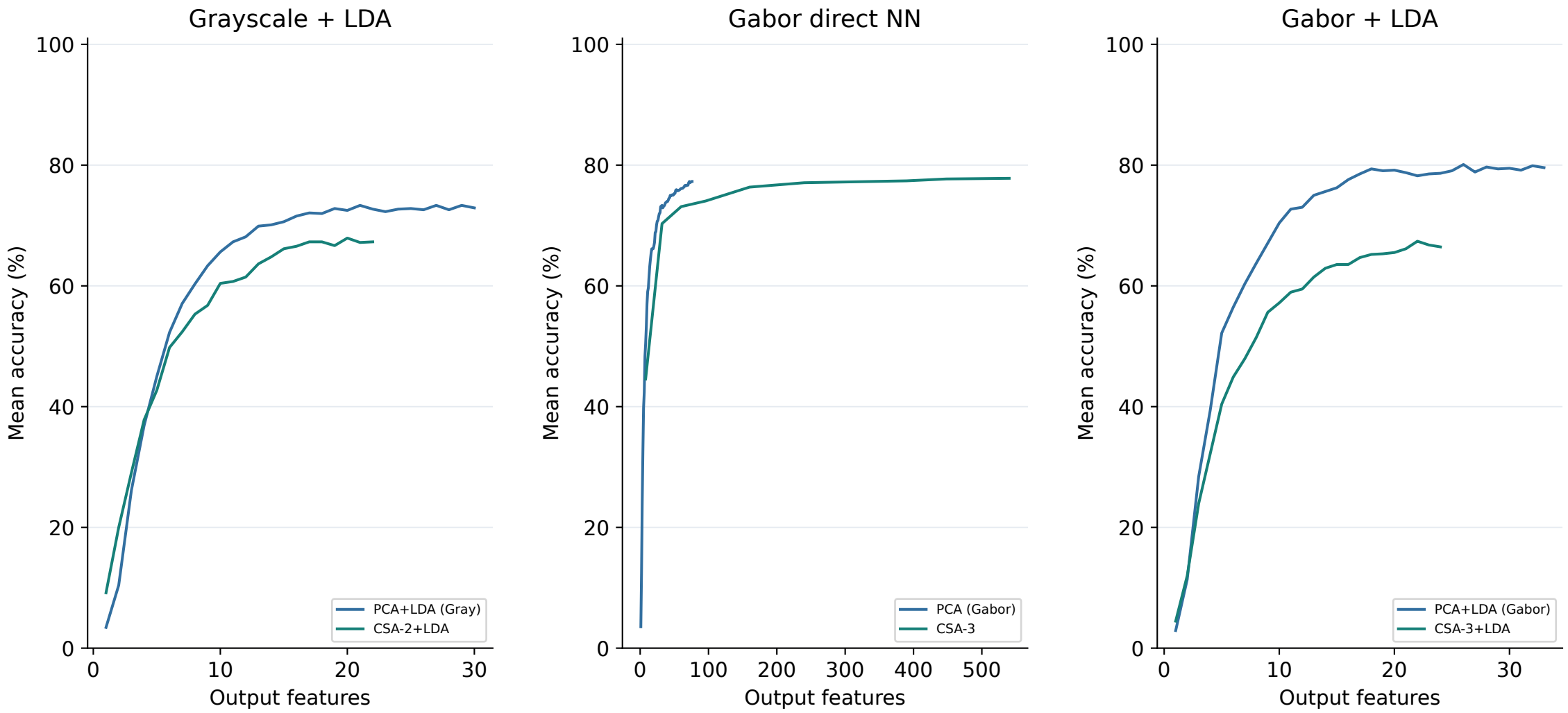
Fixed shared-cosine controls; original pre-LDA dimensions for each training size.



One fixed identity partition. These curves are descriptive evaluations across dimensions; do not reselect the winner using the held-out curve.
They correspond to our fixed control, not pointwise recovery of original Fig.6.

ORL historical dimension curves: three representation settings

Early input version and three split seeds; kept as historical recognition-curve evidence.



Historical alignment/Gabor/classification settings differ from the current88.75% HOSVD/context representative. Do not splice the current endpoint into these early curves or label all curves as one final pipeline.

CSA pipeline and the full reproduction scope

Group22 | ORL, MNIST and grayscale FERET; original PIE omitted with instructor permission.



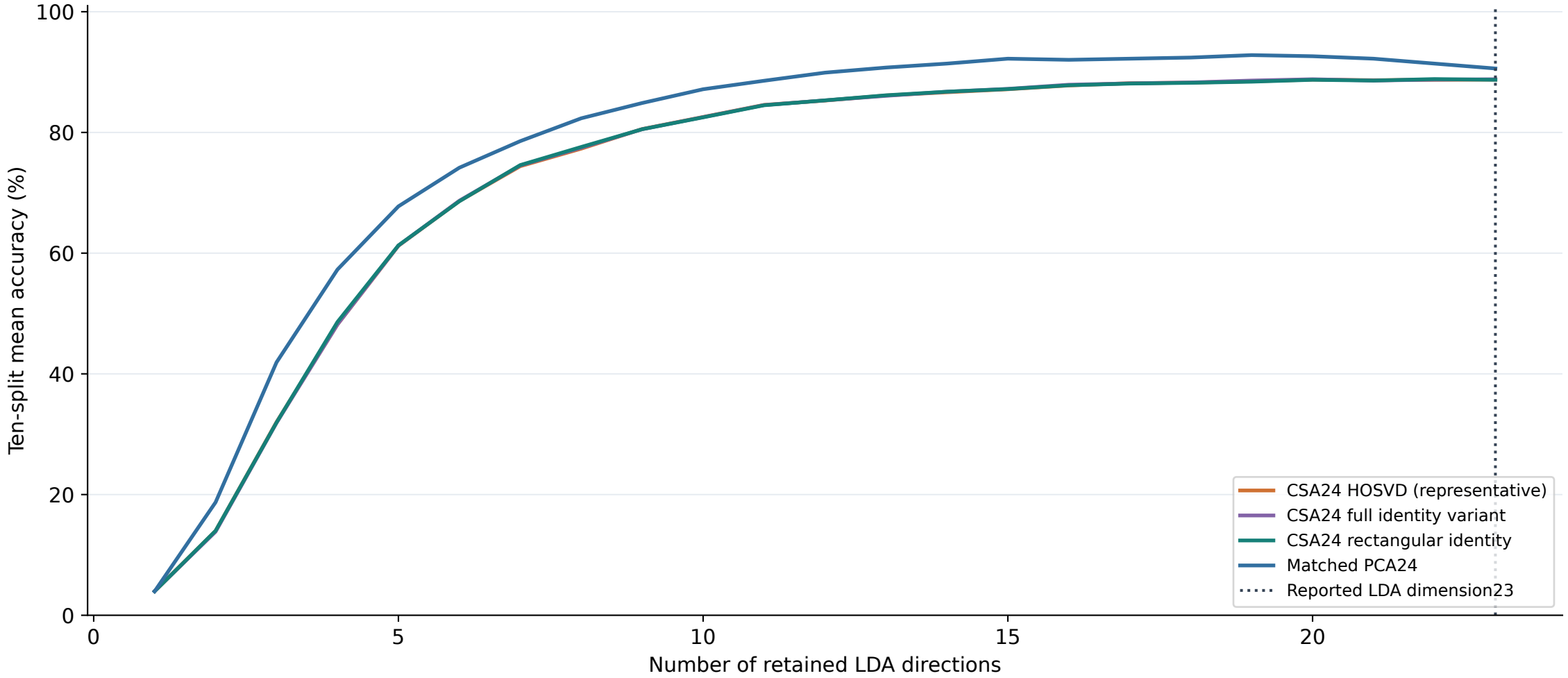
Fitting/gallery only: CSA bases, HOSVD initialization, means and LDA.
The sample axis is never compressed. Probe images do not fit the representation.

Report categories separately: shared references | gallery selections | fixed extensions
initialization variants | probe-selected extrema

Independent implementation of the paper method within the approved scope. Current ORL representative uses HOSVD initialization and context extension; full-square initialization is separate. Numerical agreement, implementation coverage and author-identical protocol are distinct claims.

ORL current input: LDA dimension curves from saved models

Zero new fits. Ten existing splits20-29; d20/y24, pi-context, 24D core, centered-cosine1-NN.



Descriptive scoring of saved directions1-23; no re-fitting or new parameter selection. Every23D endpoint matches the archived results. The fixed23D representative is retained even if a different curve point happens to score higher. Already-observed probes, context extension.